

Dynamic AI-Augmented Game-Theoretic Model of Ransomware

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Dynamic AI-Augmented Game-Theoretic Model of Ransomware

The model developed here is a culmination of several academic papers and real-world observations. Each source contributes a specific component, which is either adopted directly or extended to address limitations.

- **Laszka, Farhang & Grossklags (2017).**

Provides the game-theoretic foundation. This model adopts the two-group framework (G_1, G_2) , backup effort decision b_i , ransom payment rule $p_i = 1 \iff r \leq L_j/b_i$, attack success probability V_j , defender payoff structure, and deterrence ratio Φ with equilibrium condition $\Phi < 1$.

While the model realistically incorporates backup costs and attacker effort distributions, it is static: difficulty, cost, and effort are exogenous, and policy instruments are absent. This model retains its economic structure while replacing all static parameters with dynamic counterparts.

- **Li & Liao (2025).**

Provides the dynamic AI arms-race engine. The model adopts the attacker and defender AI evolution equations, learning rates $\eta_{a,t}$ and $\eta_{n,t}$, random training factors, and the structural asymmetry whereby attackers learn from all N targets while defenders learn only from local experience, creating a persistent offensive advantage.

This model uses these bridge equations

$$D_t = \frac{D_0}{1 + \alpha A_{a,t}}, \quad C_{B,t} = \frac{C_{B,0}}{1 + \delta A_{n,t}},$$

which translate AI capability into the parameters of Laszka's game and serve as the interface between the two frameworks. This model also introduces background growth rates μ_a and μ_n , ensuring AI capability continues to evolve even when no attacks occur.

- **Li & Liao (2022).**

Provides the double-extortion architecture, where attackers gain revenue from both ransom payments and the sale of stolen data. This model adopts the deception parameters $\delta_{e,j}$ and $\delta_{f,j}$, transaction cost C_s , and probability β_s that attackers sell data even after the victim pays.

Their static deception effectiveness

$$h = 1 - \delta_{f,j}^{1/2}$$

is replaced by

$$h(\delta_{f,j}, A_{a,t}) = 1 - \frac{\delta_{f,j}^{1/2}}{1 + \kappa_h A_{a,t}},$$

capturing how increasing attacker AI capability gradually neutralises defensive deception mechanisms. As $A_{a,t}$ grows, $h \rightarrow 1$ and defender deception becomes progressively ineffective.

- **Yin, Sarabi & Liu (2023).**

Contributes the insurance and recovery-failure components. This model adopts the moral hazard mechanism whereby insurance coverage reduces effective losses, lowering backup investment and

increasing attacker profitability. Specifically, this model retains the probabilistic recovery-failure concept, premium function

$$\pi_i(b_i) = \pi_0 e^{-\kappa_\pi b_i},$$

and indemnity fraction ξ .

The recovery failure is further endogenised as

$$\epsilon_t = \frac{\epsilon_0(1 + \kappa_\epsilon A_{a,t})}{(1 + \kappa_b b_t^*)(1 + \kappa_\epsilon^{\text{def}} A_{n,t})},$$

where attacker AI degrades backup integrity while defender AI protects it, generating the backup treadmill.

The model further introduces the insurer AI subsidy

$$\mu_n^{\text{ins}} = \mu_n + \kappa_\Delta \xi,$$

motivated by cyber insurers' threat-monitoring and incident-response services. By accelerating defender AI growth through shared threat intelligence, this subsidy counteracts moral hazard.

- **August, Dao & Niculescu (2022).**

Contributes the welfare framing and network-externality perspective. Their marginal attacker participation condition directly motivates the data-selling floor

$$\beta_s S_j^{(t)} > \frac{C_D}{|G|},$$

which acts as the profitability threshold below which attackers do not enter the market.

The attacker is also modelled as a rational agent who recalculates optimal effort a_{sum}^* every round using the first-order condition.

- **Meurs, Cartwright & Cartwright (2024).**

Demonstrate that double extortion increases paid ransoms and that victims' private information significantly shapes negotiation outcomes. This motivates the inclusion of a group-specific data market value M_j , distinct from the operational loss L_j .

The paper also motivates the interpretation of β_s , reflecting situations where attackers do not always sell stolen data despite having the opportunity to do so because of reputational considerations.

- **Real-World Motivation.**

Several real-world dynamics further shape the model. The law-enforcement takedown mechanism is motivated by the FBI and Europol operations against LockBit (2024) and Hive (2023), where cumulative attack history generated the intelligence footprint necessary for interdiction.

The obfuscation-nullification threshold, where

$$\lambda_t \rightarrow 0 \quad \text{as} \quad A_{a,t} \rightarrow \infty,$$

reflects the increasing use of AI-powered cryptocurrency mixing, jurisdiction hopping, and deep-fake impersonation by ransomware operators to evade attribution, a dynamic not formally captured in prior models.

Dynamic AI-Augmented Game-Theoretic Model of Ransomware

Framework Overview & Model Architecture

1. Strategic Players ---

1.1 The Attacker

A single ransomware operator (or unified cartel) targeting organisations for extortion. The attacker possesses a **structural learning advantage**: by engaging with the entire population of targets, its AI learns from a far richer and more diverse stream of encounters than any single defender.

The attacker pursues a double-extortion model with two concurrent revenue streams:

- **The Ransom Channel:** Encrypting data and demanding payment.
- **The Data-Selling Channel:** Selling exfiltrated data on underground markets regardless of ransom payment. This creates a *data-selling floor*, sustaining attacker incentives even if law enforcement entirely blocks ransom collections.

1.2 The Defending Organisations

Multiple independent organisations (e.g., healthcare, education) with varying data valuations and loss profiles. Defenders independently choose backup investments and ransom payment strategies. Because they do not coordinate, their individually rational choices lead to suboptimal protection for the overall population. Crucially, a defender’s AI evolves solely from its own direct encounters, resulting in a limited observational horizon that places defensive AI at a structural disadvantage over time.

2. Core Mechanisms ---

The AI Arms Race

AI capabilities evolve round-by-round based on time (diminishing returns), past operational outcomes, random shocks, and opponent strength. Because attackers face N defenders while each defender faces one attacker, offensive AI generally outpaces defensive AI in the long run.

Attack Success and the Ransom Cycle

The attacker targets organisations based on expected profit. Adaptive ransom demands adjust through repeated interactions, raising after payments and lowering after refusals, progressively zeroing in on a victim’s true willingness to pay.

Recovery Failure and the Backup Treadmill

Backup investment is not a static hedge. As offensive AI advances, it increasingly targets and surpasses backup infrastructure. Defenders are forced onto a *backup treadmill*—they must continually increase their backup expenditures simply to maintain a constant baseline of effective protection.

Deterrence and Collapse

The ecosystem’s stability is tracked by a deterrence ratio, Φ_t . When $\Phi_t < 1$, the expected costs of attacking outweigh the returns, and the attacker is deterred. Once $\Phi_t \geq 1$,

deterrence collapses and attacks commence. Policy interventions aim to maximize this safe window.

3. The Spirit of Cyber Insurance

Cyber insurance introduces two opposing forces that dictate ecosystem stability:

- **The Destabilising Channel (Moral Hazard):** When policies cover a fraction ξ of data loss, organisations rationally reduce backup efforts. This increases backup failures, raises victims’ willingness to pay ransoms, and ultimately empowers the attacker.
- **The Stabilising Channel (Insurer AI Subsidy):** To reduce claim frequencies, insurers proactively subsidise defender AI (e.g., through intelligence sharing or technical grants). Insurance only improves deterrence if this AI subsidy is strong enough to offset the moral hazard, defined by a specific *design frontier* $\kappa_{\Delta}^*(\xi)$.

4. Architecture of the Four Models

The framework is built around four nested configurations, isolating different policy instruments before combining them.

MODEL A (The Baseline)

This represents the unmitigated version of the cyber landscape. Neither cyber insurance nor law enforcement takedowns are active. It serves as the foundational reference point, capturing the pure dynamics of the AI arms race, the baseline backup failure rate, and the uncontested double-extortion data market.

MODEL B (Baseline + Insurance)

This model isolates the net effect of private-sector intervention by activating cyber insurance while keeping law enforcement takedowns strictly inactive. It focuses entirely on the internal tug-of-war between the insurance-induced moral hazard and the insurer-subsidized boost to defender AI.

MODEL C (Baseline + Law Enforcement Takedown)

This model isolates the net effect of public-sector intervention. It activates law enforcement takedowns while turning insurance off. It highlights how law enforcement can delay systemic collapse but ultimately reveals its fatal flaw: the inability to eliminate the data-selling floor and its vulnerability to AI-driven obfuscation.

MODEL D (Full Combined Synthesis)

The ultimate landscape where every extension, the AI arms race, cyber insurance, and law enforcement takedowns operate simultaneously. It explores the complex interaction between public and private interventions, ultimately questioning whether the insurer’s AI subsidy can provide a “double-dividend” that simultaneously offsets the insurer’s own moral hazard and the limitations of law enforcement takedowns.

Game Theory Models: Ransomware Framework

5. Symbols

Symbol	Meaning
$G_j, G_j $	Organisation group j and its size.
W_j	Initial wealth of victim group j.
F_j	Fixed recovery loss after attack.
L_j	Permanent long-term loss from attack.
T_j	Business interruption cost during attack.
β	Behavioural discount parameter.
b_i	Backup effort chosen by organisation i.
p_i	Binary ransom payment decision.
r^*	Equilibrium ransom demand.
C_A	Marginal attacker effort cost.
C_D	Fixed attacker development cost.
D_t	Attack difficulty in round t.
D_0	Initial attack difficulty.
$C_{B,t}$	Backup cost in period t.
$C_{B,0}$	Initial backup cost.
α	Attacker AI scaling coefficient.
δ	Defender AI scaling coefficient.
$A_{a,t}$	Attacker AI capability in period t.
$A_{n,t}$	Defender AI capability in period t.
μ_a	Baseline attacker AI growth rate.
μ_n	Baseline defender AI growth rate.
$\eta_{a,t}$	Attacker learning rate.
$\eta_{n,t}$	Defender learning rate.
$r_{a,t}$	Random attacker training factor.
$r_{n,t}$	Random defender training factor.
s_t^{op}	Attacker operational success indicator, which is 0 during deterred rounds.
f_t^{op}	Defender operational success indicator, which is 1 during deterred rounds.
Φ_t	Deterrence ratio. It is the central stability indicator of the model: deterrence holds whenever $\Phi_t < 1$, while $\Phi_t > 1$ indicates systemic deterrence collapse.
ρ_t	AI advantage ratio. This measures the relative strength of attacker AI compared to defender AI and determines which side dominates the cyber ecosystem dynamically.
M_j	Raw market value of stolen organisational data. It represents how profitable leaked information becomes on underground data markets.

Symbol	Meaning
$\delta_{e,j}$	Pre-encrypted fraction of organisational data. Higher values reduce attacker monetisation opportunities because some information is already protected.
$\delta_{f,j}$	Fraction of fake or poisoned data intentionally introduced by defenders. This variable models deception-based cyber defence strategies.
κ_h	AI erosion coefficient of deception effectiveness. Strong attacker AI gradually learns to distinguish fake data from valuable data, reducing deception efficiency over time.
$h(\delta_{f,j}, A_{a,t})$	Residual value function of stolen data after accounting for deception and attacker AI filtering capability. It determines how much of the stolen dataset remains economically useful.
C_s	Transaction cost of selling stolen data. It includes laundering costs, dark-web transaction frictions, brokerage fees, and operational risk.
β_s	Probability that attackers still sell data even when victims pay ransom. This captures the double-extortion nature of modern ransomware ecosystems.
$S_j^{(t)}$	Net data-selling profit earned from group j in round t. This creates the "data-selling floor" that sustains attacker incentives even under strong law enforcement takedown.
ϵ_0	Baseline probability that recovery systems fail after an attack. It reflects imperfect backup reliability even before AI escalation occurs.
κ_b	Backup-efficiency coefficient. It determines how strongly backup investment reduces recovery failure probability.
κ_ϵ	Attacker-AI corruption coefficient. Higher attacker AI increasingly sabotages or bypasses backup systems, increasing recovery failure risk.
κ_ϵ^{def}	Defender-AI recovery protection coefficient. It measures how effectively defender AI improves backup integrity and system resilience.
ϵ_t	Endogenous recovery failure probability at time t. This variable dynamically evolves based on attacker AI, defender AI, and backup investment quality.
ξ	Insurance indemnity fraction. It measures the percentage of ransomware losses covered by cyber insurance policies.
π_0	Baseline insurance premium level before behavioural adjustments.
κ_π	Premium sensitivity parameter. It determines how strongly insurance pricing responds to backup effort and organisational security posture.
$\pi_i(b_i)$	Per-round insurance premium paid by organisation i. This introduces a strategic interaction between insurance pricing and cyber investment incentives.
κ_Δ	Insurer AI subsidy scaling coefficient. It measures how strongly insurers accelerate defender AI investment through subsidies and technological support.
μ_n^{ins}	Insurance-enhanced defender AI growth rate. This captures how cyber insurance ecosystems indirectly accelerate cybersecurity innovation.
λ_0	Baseline law enforcement takedown probability. It measures the initial ability of authorities to disrupt ransomware operations.
κ_λ	Attacker-AI obfuscation scaling coefficient. Stronger attacker AI reduces the effectiveness of law enforcement takedown tracking and takedown operations.

Symbol	Meaning
λ_t	Endogenous takedown probability at time t. It dynamically evolves with attacker obfuscation capability and historical law enforcement takedown success.
Ψ_t	Attacker survival probability. This determines the fraction of ransomware operations that remain active after accounting for law enforcement takedown pressure.
s_{t-1}^{cum}	Rolling historical attack success rate. It serves as an adaptive feedback signal for law enforcement takedown effectiveness.

6. Equations

6.1 General Equations

$$D_t = \frac{D_0}{1 + \alpha A_{a,t}} \quad (1)$$

This shows how attack difficulty changes as attacker AI improves. Higher attacker AI reduces difficulty, making it easier to compromise targets.

$$C_{B,t} = \frac{C_{B,0}}{1 + \delta A_{n,t}} \quad (2)$$

This shows how backup cost changes as defender AI improves. Better defensive AI lowers the cost of maintaining effective backups.

$$A_{a,t} = A_{a,t-1}(1 + \mu_a)(1 + \eta_{a,t})(1 + r_{a,t}s_t^{op}) \quad (3)$$

This equation describes how attacker AI grows over time. It combines baseline progress, learning from experience, and reinforcement from operational success, so attacks can keep getting stronger when they remain profitable.

$$A_{n,t} = A_{n,t-1}(1 + \mu_n)(1 + \eta_{n,t})(1 + r_{n,t}f_t^{op}) \quad (4)$$

This shows how defender AI improves through innovation and learning. Under insurance-backed settings, this growth can speed up, which gives defenders a better chance of offsetting attacker dominance.

6.2 Attack Success Probability

$$V_j^{(t)}(a_1, a_2) = \frac{a_j^*(t)(1 + \alpha A_{a,t})}{D_0 + a_{sum}^*(t)(1 + \alpha A_{a,t})} \quad (5)$$

This captures the likelihood that attackers successfully compromise organisations in group j. The numerator reflects offensive strength amplified by attacker AI, while the denominator reflects overall ecosystem resistance and total attack pressure. Here, $a_j^*(t)$ represents the optimal attacker effort directed at victim group j, while $a_{sum}^*(t)$ is the total endogenous attacker effort across all targets.

6.3 Deception Effectiveness Function

$$h(\delta_{f,j}, A_{a,t}) = 1 - \frac{\delta_{f,j}^{1/2}}{1 + \kappa_h A_{a,t}} \quad (6)$$

This shows how much value remains in stolen data after defenders inject fake or poisoned information. At zero attacker AI, deception is maximally effective. As attacker AI grows, it learns to filter out misleading data, restoring the full market value of stolen information.

6.4 Data-Selling Profit Function

$$S_j^{(t)} = \max(0, (1 - \delta_{e,j})h(\delta_{f,j}, A_{a,t})M_j - C_s) \quad (7)$$

This gives the net profit attackers make by selling stolen data in underground markets. It captures the double-extortion side of ransomware and explains why attackers still have a revenue stream even when ransom collection weakens.

6.5 Recovery Failure Probability Without Defender-AI Coupling

$$\epsilon_t^{A,C} = \frac{\epsilon_0(1 + \kappa_\epsilon A_{a,t})}{1 + \kappa_b b_t^*} \quad (8)$$

This describes recovery failure in Models A and C (no insurer AI subsidy). Backup investment helps reduce failure risk, but stronger attacker AI pushes that risk upward, which is why organisations often need to keep increasing defensive effort just to stay level.

6.6 Recovery Failure Probability With Defender-AI Coupling

$$\epsilon_t^{B,D} = \frac{\epsilon_0(1 + \kappa_\epsilon A_{a,t})}{(1 + \kappa_b b_t^*)(1 + \kappa_\epsilon^{def} A_{n,t})} \quad (9)$$

This extends the previous case to Models B and D, where the insurer AI subsidy accelerates $A_{n,t}$, progressively protecting backup integrity. The model balances two forces: moral hazard can weaken backup effort, while insurance-backed AI improvements can strengthen resilience.

6.7 Learning Rates

$$\eta_{a,t} = \frac{1}{1 + (A_{a,t-1}/A_{n,t-1})t}, \quad \eta_{n,t} = \frac{1}{1 + (A_{n,t-1}/A_{a,t-1})t} \quad (10)$$

These equations capture how attackers and defenders learn over time. As one side becomes more dominant in the AI race, further learning becomes harder, which keeps growth from becoming unrealistically explosive.

6.8 AI Advantage Ratio

$$\rho_t = \frac{\alpha A_{a,t}}{\delta A_{n,t}} \quad (11)$$

This compares attacker AI strength with defender AI strength. Values above one mean attackers are ahead, while values below one suggest defenders currently have the upper hand.

6.9 Insurance Premium Function

$$\pi_i(b_i) = \pi_0 e^{-\kappa_\pi b_i} \quad (12)$$

This shows how cyber insurance premiums change with backup effort. Better preparation can lower premiums, which also gives insurers a reason to reward stronger security behaviour.

6.10 Insurance-Subsidised Defender AI Growth

$$\mu_n^{ins} = \mu_n + \kappa_\Delta \xi \quad (13)$$

This captures the indirect boost insurance gives to defender AI. In simple terms, insurance can help cybersecurity improve faster by supporting investment, monitoring, and defensive infrastructure.

6.11 Law enforcement takedown Probability

$$\lambda_t = \lambda_0 \cdot \frac{s_{t-1}^{cum}}{1 + \kappa_\lambda A_{a,t}} \quad (14)$$

This measures how likely law enforcement takedown is to disrupt ransomware operations. Stronger enforcement helps, but increasingly sophisticated attacker AI makes tracking and takedowns harder over time.

6.12 Attacker Survival Probability

$$\Psi_t = 1 - \lambda_t \quad (15)$$

This gives the share of ransomware operations that survive law enforcement takedown pressure. It directly affects expected ransom revenue and helps show whether enforcement can meaningfully slow down deterrence collapse.

6.13 AI Evolution (Models B and D)

$$A_{a,t} = A_{a,t-1}(1 + \mu_a)(1 + \eta_{a,t})(1 + r_{a,t}s_t^{op}) \quad (16)$$

$$A_{n,t}^{B,D} = A_{n,t-1}^{B,D}(1 + \mu_n^{ins})(1 + \eta_{n,t})(1 + r_{n,t}f_t^{op}) \quad (17)$$

These equations are used for AI evolution for both Model B and Model D. Because the insurance-subsidised growth rate replaces the baseline defender growth rate ($\mu_n^{ins} > \mu_n$), insured defenders grow faster during peacetime than uninsured ones, which actively narrows the AI asymmetry ratio. The subsidised rate partially counteracts the attacker's structural learning advantage, allowing the AI asymmetry ratio to narrow more quickly during deterred rounds.

7. Model A: Baseline Reference

7.1 Purpose & Scope

Model A is the threat-landscape baseline. It captures the full AI arms race, data-selling, and backup failure with no policy interventions active ($\xi = 0, \lambda_0 = 0$). Every other model is measured against it.

7.2 Shared Core Equations

Bridge equations (AI $\rightarrow$ economic parameters):

$$D_t = \frac{D_0}{1 + \alpha A_{a,t}}, \quad C_{B,t} = \frac{C_{B,0}}{1 + \delta A_{n,t}} \quad (18)$$

Attack success probability:

$$V_j^{(t)} = \frac{a_j^*(t)(1 + \alpha A_{a,t})}{D_0 + a_{sum}^*(t)(1 + \alpha A_{a,t})} \quad (19)$$

AI learning rates:

$$\eta_{a,t} = \frac{1}{1 + (A_{a,t-1}/A_{n,t-1})t}, \quad \eta_{n,t} = \frac{1}{1 + (A_{n,t-1}/A_{a,t-1})t} \quad (20)$$

7.3 Data-Selling

AI-dependent deception effectiveness (corrected form attacker AI erodes deception, restoring data value as $A_{a,t} \rightarrow \infty$):

$$h(\delta_{f,j}, A_{a,t}) = 1 - \frac{\delta_{f,j}^{1/2}}{1 + \kappa_h A_{a,t}} \quad (21)$$

At $A_{a,t} = 0$: $h = 1 - \delta_f^{1/2}$ (maximum deception, recovers Li & Liao 2022). As $A_{a,t} \rightarrow \infty$: $h \rightarrow 1$ (attacker perfectly filters fakes, full data value restored).

Net data-selling profit per group:

$$S_j^{(t)} = \max(0, (1 - \delta_{e,j})h(\delta_{f,j}, A_{a,t})M_j - C_s) \quad (22)$$

7.4 Recovery Failure

Endogenous backup failure probability (attacker AI neutralizes backups; more backup investment reduces failure):

$$\epsilon_t^A = \frac{\epsilon_0(1 + \kappa_\epsilon A_{a,t})}{1 + \kappa_b b_t^{*,A}} \quad (23)$$

Note 1. ϵ_t^A couples the attacker's AI evolution directly to backup reliability. Even with optimal $b_t^{*,A}$, rising $A_{a,t}$ steadily increases ϵ_t^A , creating the backup treadmill: defenders invest more each round to maintain the same effective protection level.

7.5 Defender Payoffs

Not compromised:

$$U_{nc}^{Oi,A} = W_j - C_{B,t}b_i - \beta \frac{F_j}{b_i} \quad (24)$$

Compromised:

$$U_c^{Oi,A} = W_j - C_{B,t}b_i - \beta \left[\frac{F_j + (1 - p_i)L_j}{b_i} + T_j + p_i r^{*,A} \right] \quad (25)$$

Expected utility:

$$E[U^{Oi}]^A = (1 - V_j^{(t)})U_{nc}^{Oi,A} + V_j^{(t)}U_c^{Oi,A} \quad (26)$$

7.6 Attacker Payoff

($\Psi_t = 1$, no law enforcement takedown)

$$E[U_A]^A = \sum_j |G_j| V_j^{(t)} p_i r^{*,A} + \sum_j |G_j| V_j^{(t)} [\beta_s p_i + (1 - p_i)] S_j^{(t)} - C_A a_{sum} - C_D \quad (27)$$

If victim pays ($p_i = 1$): attacker sells data with probability β_s . If victim does not pay ($p_i = 0$): attacker sells data with certainty.

7.7 Laszka Fixed-Point Sequence

Step 1 Ransom payment threshold (victim pays iff):

$$p_i = 1 \iff r^{*,A} \leq \frac{\epsilon_t^A L_j}{b_i} \quad (28)$$

Recovery failure ϵ_t^A aggravates a defender's situation : if backup is more likely to fail, victims are more willing to pay.

Step 2 Optimal backup:

$$b_t^{*,A} = \sqrt{\frac{\beta[F_j + \epsilon_t^A V_j^{(t)} L_j]}{C_{B,t}}} \quad (29)$$

Step 3 Equilibrium ransom:

$$r^{*,A}(t) = \frac{\epsilon_t^A L_j}{b_t^{*,A}} \quad (30)$$

Step 4 Endogenous attacker effort :

$$a_{sum}^{*,A}(t) = \sqrt{\frac{|G| D_t}{C_A} [r^{*,A}(t) + \beta_s S_j^{(t)}]} - D_t \quad (31)$$

Due to equal sum split we have $a_1^*(t) = a_2^*(t) = a_{sum}^{*,A}/2$.

Step 5 Attack success probability: Compute $V_j^{(t)}$ via Eq. (2) using updated $a_{sum}^{*,A}$.

7.8 Deterrence Ratio & Condition

$$\Phi_t^A = \frac{\epsilon_t^A \cdot a_{sum}^{*,A} \cdot |G_2|}{(D_t + a_{sum}^{*,A})(C_A a_{sum}^{*,A} + C_D)} \cdot \sqrt{\frac{C_{B,t}}{\beta F_2}} \quad (32)$$

Deterrence holds in round t iff $\Phi_t^A < 1$. First crossing $\Phi_t^A \geq 1$ defines T_A^* (deterrence collapse round).

AI advantage ratio:

$$\rho_t = \frac{\alpha A_{a,t}}{\delta A_{n,t}}; \quad \rho_t > 1 \Rightarrow \text{eroding}, \rho_t < 1 \Rightarrow \text{strengthening} \quad (33)$$

7.9 AI Evolution (No Insurer Subsidy)

$$A_{a,t} = A_{a,t-1}(1 + \mu_a)(1 + \eta_{a,t})(1 + r_{a,t} s_t^{op}) \quad (34)$$

$$A_{n,t} = A_{n,t-1}(1 + \mu_n)(1 + \eta_{n,t})(1 + r_{n,t} f_t^{op}) \quad (35)$$

During deterred rounds: $s_t^{op} = 0$, $f_t^{op} = 1$. Background terms $(1 + \mu_a)$, $(1 + \mu_n)$ ensure both AIs grow even in peace (Fix 1 prevents infinite-peace deadlock).

7.10 Round-by-Round Game Cycle - Model A

- **Step 1: Bridge update:** $D_t \leftarrow D_0/(1 + \alpha A_{a,t-1})$; $C_{B,t} \leftarrow C_{B,0}/(1 + \delta A_{n,t-1})$.
 - **Step 1b: Env. update:** $h(\delta_{f,j}, A_{a,t-1})$; $S_j^{(t)}$; ϵ_t^A (uses previous b_t^* iterate once).
 - **Step 2: Laszka fixed-point:** $b_t^{*,A} \rightarrow r^{*,A} \rightarrow a_{sum}^{*,A} \rightarrow V_j^{(t)} \rightarrow \Phi_t^A$.
 - **Step 3: Deterrence check:** $\Phi_t^A < 1$: deterred; $s_t^{op} = 0, f_t^{op} = 1$. Else: attack; $s_t^{op} = V_j^{(t)}, f_t^{op} = 1 - V_j^{(t)}$; record T_A^* .
 - **Step 4: AI evolution:** Update $A_{a,t}, A_{n,t}$ via Eqs. (11)-(12).
 - **Step 5: Record outputs:** $\Phi_t^A, E[U_A]^A, E[U^{Oi}]^A, \rho_t, a_{sum}^{*,A}, \epsilon_t^A, D_t, C_{B,t}, A_{a,t}, A_{n,t}, S_j^{(t)}$.
-

8. Model B: Baseline + Insurance

8.1 Purpose & Scope

Model B adds cyber insurance to the threat landscape, isolating its net effect by keeping law enforcement takedown inactive ($\Psi_t = 1$). Insurance operates through two opposing channels: a destabilizing moral hazard (indemnity reduces effective loss, lowering optimal backup) and a stabilizing insurer AI subsidy (accelerates defender AI growth, reducing backup cost and failure).

8.2 Shared Core Equations

Bridge equations:

$$D_t = \frac{D_0}{1 + \alpha A_{a,t}}, \quad C_{B,t} = \frac{C_{B,0}}{1 + \delta A_{n,t}^B} \quad (36)$$

Note: $C_{B,t}$ in Model B uses $A_{n,t}^B$ (the subsidised defender AI), so the backup cost trajectory differs from Model A even with the same formula.

Attack success probability:

$$V_j^{(t)} = \frac{a_j^*(t)(1 + \alpha A_{a,t})}{D_0 + a_{sum}^*(t)(1 + \alpha A_{a,t})} \quad (37)$$

AI learning rates:

$$\eta_{a,t} = \frac{1}{1 + (A_{a,t-1}/A_{n,t-1}^B)t}, \quad \eta_{n,t} = \frac{1}{1 + (A_{n,t-1}^B/A_{a,t-1})t} \quad (38)$$

AI advantage ratio:

$$\rho_t = \frac{\alpha A_{a,t}}{\delta A_{n,t}^B} \quad (39)$$

8.3 Data-Selling

AI-dependent deception effectiveness (corrected form):

$$h(\delta_{f,j}, A_{a,t}) = 1 - \frac{\delta_{f,j}^{1/2}}{1 + \kappa_h A_{a,t}} \quad (40)$$

As $A_{a,t} \rightarrow \infty$, $h \rightarrow 1$ (attacker fully filters fake data, restoring full market value). At $A_{a,t} = 0$: $h = 1 - \delta_f^{1/2}$ (maximum deception).

Net data-selling profit per group:

$$S_j^{(t)} = \max(0, (1 - \delta_{e,j})h(\delta_{f,j}, A_{a,t})M_j - C_s) \quad (41)$$

8.4 Recovery Failure with Defender-AI Coupling

In Model B, recovery failure has an additional denominator term absent from Models A and C: the insurer's AI subsidy accelerates $A_{n,t}^B$, which progressively protects backup integrity.

$$\epsilon_t^B = \frac{\epsilon_0(1 + \kappa_e A_{a,t})}{(1 + \kappa_b b_t^{*,B})(1 + \kappa_e^{def} A_{n,t}^B)} \quad (42)$$

Note 1 (Double-Dividend of the Insurer Subsidy). The insurer AI subsidy κ_Δ creates two simultaneous benefits: it raises $A_{n,t}^B$ directly (lowering $C_{B,t}$ via the bridge equation) and reduces ϵ_t^B through the κ_e^{def} term. This double-dividend partially offsets the moral hazard channel. Whether it dominates depends on the relative magnitudes of κ_π and κ_Δ .

8.5 Insurance Mechanism

8.5.1 Premium and AI Subsidy

$$\pi_i(b_i) = \pi_0 e^{-\kappa_\pi b_i}, \quad \mu_n^{ins} = \mu_n + \kappa_\Delta \xi \quad (43)$$

The premium π_i decreases in backup effort (insurer incentive alignment). At $\kappa_\pi = 0$ (default), the premium is flat the pure moral-hazard baseline. The subsidised growth rate μ_n^{ins} replaces μ_n in defender AI evolution for the entire game.

Note 2 (Design Frontier). The insurer can choose (ξ^*, κ_Δ^*) such that $\Phi_t^B = \Phi_t^A$ (deterrence-neutral insurance). For $\kappa_\Delta > \kappa_\Delta^*(\xi)$ the AI subsidy dominates and insurance is deterrence-improving; below it, moral hazard dominates and insurance accelerates collapse ($T_B^* < T_A^*$).

8.6 Defender Payoffs

Not compromised (premium paid regardless):

$$U_{nc}^{Oi,B} = W_j - C_{B,t}b_i - \pi_i(b_i) - \beta \frac{F_j}{b_i} \quad (44)$$

Compromised (indemnity ξ reduces effective data-loss):

$$U_c^{Oi,B} = W_j - C_{B,t}b_i - \pi_i(b_i) - \beta \left[\frac{F_j + (1 - p_i)(1 - \xi)L_j}{b_i} + T_j + p_i r^{*,B} \right] \quad (45)$$

Expected utility:

$$E[U^{Oi}]^B = (1 - V_j^{(t)})U_{nc}^{Oi,B} + V_j^{(t)}U_c^{Oi,B} \quad (46)$$

The $(1 - \xi)$ inside the compromised payoff is the sole entry point of insurance into the defender's loss calculation. It propagates upstream into $b_t^{*,B}$ and $r^{*,B}$ but cancels in Φ_t^B .

8.7 Attacker Payoff

($\Psi_t = 1$, no law enforcement takedown)

$$E[U_A]^B = \sum_j |G_j| V_j^{(t)} p_i r^{*,B} + \sum_j |G_j| V_j^{(t)} [\beta_s p_i + (1 - p_i)] S_j^{(t)} - C_A a_{sum} - C_D \quad (47)$$

At equilibrium $p_i = 1$, the data-selling term reduces to $\beta_s S_j^{(t)}$. Insurance does not enter the attacker payoff directly it affects the attacker only indirectly through the lower $r^{*,B}$ (victims pay less because their effective loss is reduced).

8.8 Laszka Fixed-Point Sequence

Step 1 Ransom payment threshold (insurance reduces effective loss):

$$p_i = 1 \iff r^{*,B} \leq \frac{\epsilon_t^B (1 - \xi) L_j}{b_i} \quad (48)$$

Compared to Model A threshold $\epsilon_t^A L_j / b_i$ the $(1 - \xi)$ factor strictly lowers the maximum ransom a victim will accept, constraining the attacker's revenue. This is the direct beneficial effect of insurance on the ransom channel.

Step 2 Optimal backup (moral hazard visible here):

$$b_t^{*,B} = \sqrt{\frac{\beta [F_j + \epsilon_t^B V_j^{(t)} (1 - \xi) L_j]}{C_{B,t}}} \quad (49)$$

Step 3 Equilibrium ransom:

$$r^{*,B}(t) = \frac{\epsilon_t^B (1 - \xi) L_j}{b_t^{*,B}} \quad (50)$$

Step 4 Endogenous attacker effort (corrected: β_s restored):

$$a_{sum}^{*,B}(t) = \sqrt{\frac{|G| D_t}{C_A} [r^{*,B}(t) + \beta_s S_j^{(t)}]} - D_t \quad (51)$$

Due to equal sum split we have $a_1^*(t) = a_2^*(t) = a_{sum}^{*,B}/2$.

Step 5 Attack success probability: Compute $V_j^{(t)}$ via Eq. (2) using updated $a_{sum}^{*,B}$.

8.9 Deterrence Ratio & Condition

$$\Phi_t^B = \frac{\epsilon_t^B \cdot a_{sum}^{*,B} \cdot |G_2|}{(D_t + a_{sum}^{*,B})(C_A a_{sum}^{*,B} + C_D)} \cdot \sqrt{\frac{C_{B,t}}{\beta F_2}} \quad (52)$$

Deterrence holds in round t iff $\Phi_t^B < 1$. First crossing $\Phi_t^B \geq 1$ defines T_B^* .

Note 2 ($(1 - \xi)$ Cancellation). The raw deterrence ratio for Model B carries $(1 - \xi)$ in both numerator and denominator. These cancel exactly:

$$\Phi_t^B = \frac{(1 - \xi) L_2 \epsilon_t^B \cdot a_{sum}^{*,B} \cdot |G_2|}{(1 - \xi) L_2 (D_t + a_{sum}^{*,B})(C_A a_{sum}^{*,B} + C_D)} \cdot \sqrt{\frac{C_{B,t}}{\beta F_2}} \quad (53)$$

The insurance effect is therefore not visible in the formula for Φ_t^B itself it has already been absorbed into ϵ_t^B , $a_{sum}^{*,B}$, $C_{B,t}$, and $A_{n,t}^B$. This structural unification confirms that the deterrence ratio formula is identical across all four models; the models differ only in the upstream quantities fed into it.

8.10 AI Evolution (Insurer-Subsidised Defender)

Attacker AI (unchanged from Model A):

$$A_{a,t} = A_{a,t-1}(1 + \mu_a)(1 + \eta_{a,t})(1 + r_{a,t}s_t^{op}) \quad (54)$$

Defender AI (subsidised growth rate): where $\mu_n^{ins} = \mu_n + \kappa_\Delta \xi > \mu_n$

$$A_{n,t}^B = A_{n,t-1}^B(1 + \mu_n^{ins})(1 + \eta_{n,t})(1 + r_{n,t}f_t^{op}) \quad (55)$$

During deterred rounds: $s_t^{op} = 0, f_t^{op} = 1$. Background terms prevent the infinite-peace deadlock (Fix 1). Because $\mu_n^{ins} > \mu_n$, insured defenders grow faster during peace than uninsured ones relative to Model A, even without any attacks occurring.

8.11 Round-by-Round Game Cycle - Model B

- **Step 1: Bridge update:** $D_t \leftarrow D_0/(1 + \alpha A_{a,t-1})$; $C_{B,t} \leftarrow C_{B,0}/(1 + \delta A_{n,t-1}^B)$.
 - **Step 1b: Ins. + Env. update:** $\pi_i(b_i) = \pi_0 e^{-\kappa_\pi b_i}$; $\mu_n^{ins} = \mu_n + \kappa_\Delta \xi$; $h(\cdot)$; $S_j^{(t)}$; ϵ_t^B .
 - **Step 2: Laszka fixed-point:** $b_t^{*,B} \rightarrow r^{*,B} \rightarrow a_{sum}^{*,B} \rightarrow V_j^{(t)} \rightarrow \Phi_t^B$.
 - **Step 3: Deterrence check:** $\Phi_t^B < 1$: deterred; $s_t^{op} = 0, f_t^{op} = 1$. Else: attack: $s_t^{op} = V_j^{(t)}, f_t^{op} = 1 - V_j^{(t)}$; record T_B^* .
 - **Step 4: AI evolution:** $A_{a,t}$ standard; $A_{n,t}^B$ with μ_n^{ins} .
 - **Step 5: Record outputs:** $\Phi_t^B, E[U_A]^B, E[U^{Oi}]^B, \rho_t, a_{sum}^{*,B}, \pi_i, \mu_n^{ins}, \epsilon_t^B, D_t, C_{B,t}, A_{a,t}, A_{n,t}^B, S_j^{(t)}$.
-

9. Model C: Baseline + Law Enforcement Takedown

9.1 Purpose & Scope

Model C adds law enforcement takedown to the threat landscape while insurance remains inactive ($\xi = 0, \Psi_t < 1$). Takedown only scales ransom revenue and cannot reach the data market, creating a strictly positive data-selling floor that sustains attacker incentives regardless of enforcement strength.

9.2 Shared Core Equations

Bridge equations:

$$D_t = \frac{D_0}{1 + \alpha A_{a,t}}, \quad C_{B,t} = \frac{C_{B,0}}{1 + \delta A_{n,t}^B} \quad (56)$$

Attack success probability:

$$V_j^{(t)} = \frac{a_j^*(t)(1 + \alpha A_{a,t})}{D_0 + a_{sum}^*(t)(1 + \alpha A_{a,t})} \quad (57)$$

AI learning rates:

$$\eta_{a,t} = \frac{1}{1 + (A_{a,t-1}/A_{n,t-1})t}, \quad \eta_{n,t} = \frac{1}{1 + (A_{n,t-1}/A_{a,t-1})t} \quad (58)$$

AI advantage ratio:

$$\rho_t = \frac{\alpha A_{a,t}}{\delta A_{n,t}} \quad (59)$$

9.3 Data-Selling

AI-dependent deception effectiveness (corrected form):

$$h(\delta_{f,j}, A_{a,t}) = 1 - \frac{\delta_{f,j}^{1/2}}{1 + \kappa_h A_{a,t}} \quad (60)$$

At $A_{a,t} = 0$: $h = 1 - \delta_f^{1/2}$ (maximum deception). As $A_{a,t} \rightarrow \infty$ $h \rightarrow 1$ (attacker fully filters fake data, full market value restored).

Net data-selling profit per group:

$$S_j^{(t)} = \max(0, (1 - \delta_{e,j})h(\delta_{f,j}, A_{a,t})M_j - C_s) \quad (61)$$

9.4 Recovery Failure (No Defender-AI Coupling)

Model C uses the same failure equation as Model A. There is no insurer AI subsidy ($\kappa_\epsilon^{def} = 0$), so the defender's AI does not protect backup integrity.

$$\epsilon_t^C = \frac{\epsilon_0(1 + \kappa_\epsilon A_{a,t})}{1 + \kappa_b b_t^{*,C}} \quad (62)$$

Rising attacker AI $A_{a,t}$ drives ϵ_t^C upward, forcing the defender onto the backup treadmill (more investment per round for the same effective protection). Unlike Model B, there is no mechanism here to push back against this erosion.

9.5 Law Enforcement Mechanism

Endogenous takedown probability:

$$\lambda_t = \lambda_0 \cdot \frac{s_{t-1}^{cum}}{1 + \kappa_\lambda A_{a,t}}, \quad \Psi_t = 1 - \lambda_t \quad (63)$$

λ_t grows with cumulative attack success s_{t-1}^{cum} (more visible attacks draw more law enforcement takedown attention) but is suppressed by attacker AI $A_{a,t}$ (better obfuscation, harder attribution, jurisdiction hopping).

Note 1 (Obfuscation Nullification Threshold). As $A_{a,t} \rightarrow \infty$, $\lambda_t \rightarrow 0$ regardless of how many successful attacks have occurred. There exists a threshold beyond which law enforcement takedown is effectively neutralised. This defines the law-enforcement horizon: the round beyond which law enforcement takedown investment provides negligible deterrence benefit.

9.6 Defender Payoffs (Corrected)

Not compromised (identical to Model A):

$$U_{nc}^{Oi,C} = W_j - C_{B,t}b_i - \beta \frac{F_j}{b_i} \quad (64)$$

Compromised (corrected: ϵ_t^C applied to non-payment penalty):

$$U_c^{Oi,C} = W_j - C_{B,t}b_i - \beta \left[\frac{F_j + (1 - p_i)\epsilon_t^C L_j}{b_i} + T_j + p_i r^{*,C} \right] \quad (65)$$

Note 2 (Why ϵ_t^C Must Appear Here). The original documentation had $(1 - p_i)L_j/b_i$ implying the victim suffers the full permanent loss L_j with certainty if they refuse to pay. This contradicts the recovery-failure premise: the victim only loses L_j if backup fails, which occurs with probability ϵ_t^C . The corrected term is the only form consistent with the payment threshold $p_i = 1 \iff r^{*,C} \leq \epsilon_t^C L_j/b_i$.

Expected utility:

$$E[U^{Oi}]^C = (1 - V_j^{(t)})U_{nc}^{Oi,C} + V_j^{(t)}U_c^{Oi,C} \quad (66)$$

9.7 Attacker Payoff (law enforcement takedown Scales Ransom Revenue Only)

$$E[U_A]^C = \Psi_t \sum_j |G_j| V_j^{(t)} p_i r^{*,C} + \sum_j |G_j| V_j^{(t)} [\beta_s p_i + (1 - p_i)] S_j^{(t)} - C_A a_{sum} - C_D \quad (67)$$

Note 3 (Law enforcement takedown Asymmetry). Ψ_t multiplies ransom revenue only. The data-selling term is outside the bracket because law enforcement takedown seizure of ransom payments does not prevent the attacker from selling already-exfiltrated data. This asymmetry is the source of the data-selling floor.

9.8 Laszka Fixed-Point Sequence

Step 1 Ransom payment threshold (no insurance, same as Model A):

$$p_i = 1 \iff r^{*,C} \leq \frac{\epsilon_t^C L_j}{b_i} \quad (68)$$

Step 2 Optimal backup (no insurance, same form as Model A):

$$b_t^{*,C} = \sqrt{\frac{\beta[F_j + \epsilon_t^C V_j^{(t)} L_j]}{C_{B,t}}} \quad (69)$$

Law enforcement takedown does not affect the defender's backup decision. The defender optimises against the attack success probability and loss magnitudes alone.

Step 3 Equilibrium ransom:

$$r^{*,C}(t) = \frac{\epsilon_t^C L_j}{b_t^{*,C}} \quad (70)$$

Step 4 Endogenous attacker effort :

$$a_{sum}^{*,C}(t) = \sqrt{\frac{|G|D_t}{C_A} [\Psi_t r^{*,C}(t) + \beta_s S_j^{(t)}]} - D_t \quad (71)$$

Due to equal sum split we have $a_1^*(t) = a_2^*(t) = a_{sum}^{*,C}/2$.

Step 5 Attack success probability: Compute $V_j^{(t)}$ via Eq. (2) using updated $a_{sum}^{*,C}$.

9.9 Deterrence Ratio & Condition

The numerator must carry Ψ_t to reflect that the attacker's expected ransom revenue is scaled by law enforcement takedown survival probability:

$$\Phi_t^C = \frac{\Psi_t \cdot \epsilon_t^C \cdot a_{sum}^{*,C} \cdot |G_2|}{(D_t + a_{sum}^{*,C})(C_A a_{sum}^{*,C} + C_D)} \cdot \sqrt{\frac{C_{B,t}}{\beta F_2}} \quad (72)$$

Deterrence holds in round t iff $\Phi_t^C < 1$. First crossing $\Phi_t^C \geq 1$ defines T_C^* .

Note 4 (Effect of Ψ_t). Since $\Psi_t \leq 1$, we have $\Phi_t^C \leq \Phi_t^A$ for all t (law enforcement takedown weakly reduces the deterrence ratio, delaying collapse). Equality holds only when $\lambda_t = 0$ (no law enforcement takedown). As attacker AI grows and $\lambda_t \rightarrow 0$ (obfuscation nullification), $\Phi_t^C \rightarrow \Phi_t^A$, confirming that law enforcement takedown loses its deterrence benefit at high AI levels.

9.10 AI Evolution (Standard, No Subsidy)

Attacker AI:

$$A_{a,t} = A_{a,t-1}(1 + \mu_a)(1 + \eta_{a,t})(1 + r_{a,t}s_t^{op}) \quad (73)$$

Defender AI (standard μ_n , no insurer subsidy):

$$A_{n,t} = A_{n,t-1}(1 + \mu_n)(1 + \eta_{n,t})(1 + r_{n,t}f_t^{op}) \quad (74)$$

During deterred rounds: $s_t^{op} = 0, f_t^{op} = 1$. Both AIs still grow via background terms. Because μ_n^{ins} is absent, the AI asymmetry ratio ρ_t follows the same trajectory as Model A, unlike Model B where the insurer subsidy narrows the gap.

9.11 Round-by-Round Game Cycle - Model C

- **Step 1: Bridge update:** $D_t \leftarrow D_0/(1 + \alpha A_{a,t-1})$; $C_{B,t} \leftarrow C_{B,0}/(1 + \delta A_{n,t-1})$.
- **Step 1b: Law enforcement takedown + Env. update:** s_{t-1}^{cum} rolling mean of s^{op} ; λ_t ; $\Psi_t = 1 - \lambda_t$; $h(\cdot)$; $S_j^{(t)}$; ϵ_t^C .
- **Step 2: Laszka fixed-point:** $b_t^{*,C} \rightarrow r^{*,C} \rightarrow a_{sum}^{*,C} \rightarrow V_j^{(t)} \rightarrow \Phi_t^C$.
- **Step 3: Deterrence check:** $\Phi_t^C < 1$: deterred; $s_t^{op} = 0, f_t^{op} = 1$. Else: attack; $s_t^{op} = V_j^{(t)}, f_t^{op} = 1 - V_j^{(t)}$; record T_C^* .
- **Step 4: AI evolution:** Update $A_{a,t}, A_{n,t}$ standard (no subsidy).
- **Step 5: Record outputs:** $\Phi_t^C, E[U_A]^C, E[U^{Oi}]^C, a_{sum}^{*,C}, \epsilon_t^C, D_t, C_{B,t}, A_{a,t}, A_{n,t}, \lambda_t, \Psi_t, s_{t-1}^{cum}, S_j^{(t)}$.

10. Model D: Full Combined

10.1 Purpose & Scope

Model D is the full combined model where every extension and policy instrument is active simultaneously. It synthesises insurance indemnity, insurer AI subsidy, law enforcement takedown, data-selling, recovery failure, the AI arms race, and bridge equations into a single unified framework.

10.2 Shared Core Equations

Bridge equations (uses $A_{n,t}^D$ the subsidised defender AI):

$$D_t = \frac{D_0}{1 + \alpha A_{a,t}}, \quad C_{B,t} = \frac{C_{B,0}}{1 + \delta A_{n,t}^D} \quad (75)$$

Attack success probability:

$$V_j^{(t)} = \frac{a_j^*(t)(1 + \alpha A_{a,t})}{D_0 + a_{sum}^*(t)(1 + \alpha A_{a,t})} \quad (76)$$

AI learning rates and AI advantage ratio:

$$\eta_{a,t} = \frac{1}{1 + (A_{a,t-1}/A_{n,t-1}^D)t}, \quad \eta_{n,t} = \frac{1}{1 + (A_{n,t-1}^D/A_{a,t-1})t} \quad (77)$$

$$\rho_t = \frac{\alpha A_{a,t}}{\delta A_{n,t}^D} \quad (78)$$

10.3 Data-Selling

AI-dependent deception effectiveness (corrected form):

$$h(\delta_{f,j}, A_{a,t}) = 1 - \frac{\delta_{f,j}^{1/2}}{1 + \kappa_h A_{a,t}} \quad (79)$$

Net data-selling profit per group:

$$S_j^{(t)} = \max(0, (1 - \delta_{e,j})h(\delta_{f,j}, A_{a,t})M_j - C_s) \quad (80)$$

10.4 Recovery Failure (Full Coupling)

Model D uses the full double-denominator form from Model B, with the critical requirement that $A_{n,t}^D$ (subsidised) is used throughout.

$$\epsilon_t^D = \frac{\epsilon_0(1 + \kappa_\epsilon A_{a,t})}{(1 + \kappa_b b_t^{*,D})(1 + \kappa_\epsilon^{def} A_{n,t}^D)} \quad (81)$$

Note 1 (Double-Dividend of Insurance in Model D). The insurer AI subsidy κ_Δ creates two compounding benefits specific to Model D: (1) it raises $A_{n,t}^D$ faster than in Models A or C, lowering $C_{B,t}$ via the bridge equation; (2) it reduces ϵ_t^D through $\kappa_\epsilon^{def} A_{n,t}^D$ in the denominator. This double-dividend partially offsets both moral hazard (which lowers $b_t^{*,D}$) and law enforcement takedown erosion. Models A and C have neither dividend; Model B has both but without law enforcement takedown interaction.

Implementation dependency chain :

- **1. Subsidised growth rate:** $\mu_n^{ins} = \mu_n + \kappa_\Delta \xi$
- **2. Advance defender AI:** $A_{n,t}^D = A_{n,t-1}^D(1 + \mu_n^{ins})(1 + \eta_{n,t})(1 + r_{n,t} f_t^{op})$
- **3. Coupled failure probability:** $\epsilon_t^D = \epsilon_0(1 + \kappa_\epsilon A_{a,t}) / [(1 + \kappa_b b_t^{*,D})(1 + \kappa_\epsilon^{def} A_{n,t}^D)]$

If $A_{n,t}^D$ is replaced by $A_{n,t}$ (unsubsidised) in step 3, Model D degenerates to Model C plus insurance moral hazard, and the double-dividend mechanism the key differentiating result of Model D will not appear in simulation output.

10.5 Insurance and law enforcement takedown (Both Active)

Insurance premium and subsidised AI growth:

$$\mu_n^{ins} = \mu_n + \kappa_\Delta \xi, \quad \pi_i(b_i) = \pi_0 e^{-\kappa_\pi b_i} \quad (82)$$

Law enforcement takedown and survival probability:

$$\lambda_t = \lambda_0 \cdot \frac{s_{t-1}^{cum}}{1 + \kappa_\lambda A_{a,t}}, \quad \Psi_t = 1 - \lambda_t \quad (83)$$

Note 2 (Instrument Interaction). Insurance and law enforcement takedown operate on different channels. Insurance affects the defender side: ξ lowers $b_t^{*,D}$ (moral hazard) while κ_Δ raises $A_{n,t}^D$ (AI subsidy). Law enforcement takedown affects the attacker side: $\Psi_t < 1$ lowers expected ransom revenue. Neither instrument directly affects the other's mechanism their interaction appears only through the shared deterrence ratio Φ_t^D .

10.6 Defender Payoffs (Corrected)

Not compromised:

$$U_{nc}^{Oi,D} = W_j - C_{B,t} b_i - \pi_i(b_i) - \beta \frac{F_j}{b_i} \quad (84)$$

Compromised (corrected: ϵ_t^D applied to non-payment penalty, consistent with payment threshold):

$$U_c^{Oi,D} = W_j - C_{B,t} b_i - \pi_i(b_i) - \beta \left[\frac{F_j + (1 - p_i) \epsilon_t^D (1 - \xi) L_j}{b_i} + T_j + p_i r^{*,D} \right] \quad (85)$$

Note 3 (Why Both ϵ_t^D and $(1 - \xi)$ Must Appear). The non-payment expected loss has two modifiers: $(1 - \xi)$ reduces the effective permanent loss because insurance covers fraction ξ of the data-compromise damage; ϵ_t^D further scales this loss because the permanent loss is only realised if the backup fails. Without ϵ_t^D , the payoff implies certain backup failure, contradicting Recovery Failure. Without $(1 - \xi)$, the payoff ignores the insurance indemnity, contradicting Insurance. Both must appear together in Model D.

Expected utility:

$$E[U^{Oi}]^D = (1 - V_j^{(t)}) U_{nc}^{Oi,D} + V_j^{(t)} U_c^{Oi,D} \quad (86)$$

10.7 Attacker Payoff (All Channels)

$$E[U_A]^D = \Psi_t \sum_j |G_j| V_j^{(t)} p_i r^{*,D} + \sum_j |G_j| V_j^{(t)} [\beta_s p_i + (1 - p_i)] S_j^{(t)} - C_A a_{sum} - C_D \quad (87)$$

Ψ_t appears on the ransom term only (law enforcement takedown-scaled), and data-selling remains law enforcement takedown-immune. Insurance affects this payoff solely through $r^{*,D}$ (lower because $(1 - \xi)$ reduces victim willingness to pay).

10.8 Laszka Fixed-Point Sequence

Step 1 Ransom payment threshold (both ϵ_t^D and $(1 - \xi)$ active):

$$p_i = 1 \iff r^{*,D} \leq \frac{\epsilon_t^D (1 - \xi) L_j}{b_i} \quad (88)$$

This is the most restrictive threshold across all four models: ϵ_t^D and $(1 - \xi)$ both reduce the maximum ransom the defender will accept.

Step 2 Optimal backup (moral hazard from ξ , recovery-failure coupling from ϵ_t^D):

$$b_t^{*,D} = \sqrt{\frac{\beta[F_j + \epsilon_t^D V_j^{(t)}(1 - \xi)L_j]}{C_{B,t}}} \quad (89)$$

Step 3 Equilibrium ransom:

$$r^{*,D}(t) = \frac{\epsilon_t^D (1 - \xi) L_j}{b_t^{*,D}} \quad (90)$$

Step 4 Endogenous attacker effort:

$$a_{sum}^{*,D}(t) = \sqrt{\frac{|G|D_t}{C_A}[\Psi_t r^{*,D}(t) + \beta_s S_j^{(t)}]} - D_t \quad (91)$$

Due to equal sum split we have $a_1^*(t) = a_2^*(t) = a_{sum}^{*,D}/2$.

Step 5 Attack success: Compute $V_j^{(t)}$ via Eq. (2) with updated $a_{sum}^{*,D}$.

10.9 Deterrence Ratio & Condition

$$\Phi_t^D = \frac{\Psi_t \cdot \epsilon_t^D \cdot a_{sum}^{*,D} \cdot |G_2|}{(D_t + a_{sum}^{*,D})(C_A a_{sum}^{*,D} + C_D)} \cdot \sqrt{\frac{C_{B,t}}{\beta F_2}} \quad (92)$$

Deterrence holds in round t iff $\Phi_t^D < 1$. First crossing $\Phi_t^D \geq 1$ defines T_D^* .

Note 4 ($(1 - \xi)$ Cancellation in Model D). The raw deterrence ratio for Model D carries $(1 - \xi)$ in both numerator and denominator. These cancel exactly (as in Model B):

$$\Phi_t^D = \frac{\Psi_t \cdot (1 - \xi) L_2 \epsilon_t^D a_{sum}^{*,D} |G_2|}{(1 - \xi) L_2 (D_t + a_{sum}^{*,D})(C_A a_{sum}^{*,D} + C_D)} \cdot \sqrt{\frac{C_{B,t}}{\beta F_2}} \quad (93)$$

Also, the redundant L_2 in the denominator and numerator cancels as well when both instances are present. The final formula above is the simplified, structurally clean form. Insurance effects enter through ϵ_t^D , $b_t^{*,D}$, $C_{B,t}$, $A_{n,t}^D$; law enforcement takedown enters through Ψ_t .

10.10 AI Evolution (Insurer-Subsidised Defender)

Attacker AI (unchanged from all models):

$$A_{a,t} = A_{a,t-1}(1 + \mu_a)(1 + \eta_{a,t})(1 + r_{a,t} s_t^{op}) \quad (94)$$

Defender AI (subsidised unique to Models B and D): where $\mu_n^{ins} = \mu_n + \kappa_\Delta \xi > \mu_n$.

$$A_{n,t}^D = A_{n,t-1}^D(1 + \mu_n^{ins})(1 + \eta_{n,t})(1 + r_{n,t} f_t^{op}) \quad (95)$$

During deterred rounds: $s_t^{op} = 0$, $f_t^{op} = 1$. Background growth prevents peace deadlock (Fix 1). Because $\mu_n^{ins} > \mu_n$, the AI asymmetry ratio ρ_t narrows more quickly in Model D than in Model C, partially counteracting the attacker's structural learning advantage.

10.11 Round-by-Round Game Cycle - Model D

- **Step 1: Bridge update:** $D_t \leftarrow D_0/(1 + \alpha A_{a,t-1})$; $C_{B,t} \leftarrow C_{B,0}/(1 + \delta A_{n,t-1}^D)$.
- **Step 1b: Full update:** (i) π_i, μ_n^{ins} (ii) $\lambda_t, \Psi_t, s_{t-1}^{cum}$ (iii) $h(\cdot), S_j^{(t)}$; (iv) ϵ_t^D using $A_{n,t}^D$.
- **Step 2: Laszka fixed-point:** $b_t^{*,D} \rightarrow r^{*,D} \rightarrow a_{sum}^{*,D} \rightarrow V_j^{(t)} \rightarrow \Phi_t^D$.
- **Step 3: Deterrence check:** $\Phi_t^D < 1$: deterred; $s_t^{op} = 0, f_t^{op} = 1$. Else: attack; $s_t^{op} = V_j^{(t)}, f_t^{op} = 1 - V_j^{(t)}$; record T_D^* .
- **Step 4: AI evolution:** $A_{a,t}$ standard; $A_{n,t}^D$ with μ_n^{ins} .
- **Step 5: Record outputs:** $\Phi_t^D, E[U_A]^D, E[U^{O_i}]^D, \rho_t, a_{sum}^{*,D}, \epsilon_t^D, D_t, C_{B,t}, A_{a,t}, A_{n,t}^D, \lambda_t, \Psi_t, S_j^{(t)}, \pi_i, \mu_n^{ins}$.

Note 5 (Step 1b Ordering is Mandatory). Sub-steps (i)-(iv) within Step 1b must execute in sequence. μ_n^{ins} must be set before advancing $A_{n,t}^D$; $A_{n,t}^D$ must be computed before ϵ_t^D . Reversing this order breaks the double-dividend coupling and makes Model D output identical to Model C, eliminating the key distinguishing result.

Dynamic AI-Augmented Game-Theoretic Model of Ransomware

Mathematical Proofs and Results

Preliminaries

The model is within the unified framework bridging Laszka et al. (2017) and Li & Liao (2025), extended with data-selling, recovery failure, cyber insurance, and law enforcement. The deterrence ratio for each model, after algebraic simplification (cancellation of L_2 and $(1 - \xi)$ where applicable), admits the common structural form

$$\Phi_t = \Psi_t \cdot \frac{\epsilon_t \cdot a_{\text{sum}}^* \cdot |G_2|}{(D_t + a_{\text{sum}}^*)(C_A a_{\text{sum}}^* + C_D)} \cdot \sqrt{\frac{C_{B,t}}{\beta F_2}}, \quad (1)$$

where Ψ_t , ϵ_t , a_{sum}^* , and $C_{B,t}$ are model-specific. The endogenous attacker effort under the cartel assumption satisfies

$$a_{\text{sum}}^*(t) = \sqrt{\frac{|G|D_t}{C_A} [\Psi_t r^*(t) + \beta_s S_j^{(t)}]} - D_t, \quad (2)$$

with $\Psi_t = 1$ for Models A and B. Deterrence holds in round t if and only if $\Phi_t < 1$. The deterrence collapse time for model $X \in \{A, B, C, D\}$ is defined as

$$T_X^* = \min\{t \geq 1 : \Phi_t^X \geq 1\}. \quad (3)$$

The following results from Laszka et al. (2017) are invoked throughout.

Proposition 1 (Laszka et al., Proposition 1). *The no-attack equilibrium ($a_1^* = a_2^* = 0$) with optimal backup $\hat{b}_j^* = \sqrt{\beta F_j / C_B}$ is a Nash equilibrium if and only if $\Phi < 1$.*

Proposition 2 (Laszka et al., Lemma 1). *Organisation $i \in G_j$ pays the ransom if and only if $r \leq L_j/b_i$. In the extended framework this generalises to $r \leq \epsilon_t(1 - \xi)L_j/b_i$.*

Assumptions

Assumption 1 (Attacker AI dominance). $\alpha > \delta$ and $\mu_a > \mu_n$, so that attacker AI capability grows strictly faster than defender AI in expectation.

Assumption 2 (Interior equilibrium). *Parameters are in the interior region such that all four models exhibit finite $T^* < T$ and all relevant equilibrium segments are present over some sub-interval of rounds.*

Assumption 3 (Moral hazard dominance). *The insurer AI subsidy is sufficiently weak relative to the indemnity fraction that the backup-reducing moral hazard channel of insurance dominates its AI-subsidy stabilisation channel. The precise threshold separating these two regimes is characterised in Result 1 below; Assumption 3 stipulates that parameters lie strictly below it.*

Assumption 4 (Non-trivial data market). *The net data-selling profit satisfies $\beta_s S_j^{(t)} > 0$ for all rounds in the relevant horizon and $\beta_s S_j^{(t)} > C_D/|G|$ for all sufficiently large t .*

Assumption 5 (Positive prior attack history). *The initial cumulative attack success rate satisfies $s_0^{\text{cum}} > 0$, so law enforcement exerts non-trivial pressure from round $t = 1$.*

Auxiliary Lemmas

Lemma 1 (Monotonicity of Φ_t in time). *Under Assumption 1, Φ_t^X is eventually strictly increasing in t for each model $X \in \{A, B, C, D\}$.*

Proof. Under Assumption 1, the bridge equation $D_t = D_0/(1 + \alpha A_{a,t})$ gives $D_t \rightarrow 0$ since $\alpha A_{a,t} \rightarrow \infty$. The recovery failure numerator $(1 + \kappa_\epsilon A_{a,t})$ grows faster than the denominator under the backup treadmill condition $\kappa_\epsilon \mu_a > \kappa_b \mu_n$, implied by Assumption 1 and standard parametrisation, so $\epsilon_t \uparrow$. As $D_t \rightarrow 0$, Eq. (2) gives $a_{\text{sum}}^* \rightarrow 0$ yet $a_{\text{sum}}^*/D_t \rightarrow \infty$, and the infection probability $V_j^{(t)} \rightarrow 1$. Since ϵ_t and V_j are eventually increasing and dominate the denominator terms in Eq. (1), Φ_t is eventually strictly increasing in t . $\square$

Lemma 2 (Uniqueness of T^*). *Under Assumptions 1–2 and Lemma 1, T_X^* is well-defined and unique for each model X .*

Proof. Assumption 2 gives $\Phi_0^X < 1$ and $\Phi_T^X > 1$ for some finite T . By Lemma 1, Φ_t^X is eventually strictly increasing, so it crosses $\Phi = 1$ exactly once from below. The minimum in Eq. (3) is attained at a unique T_X^* . $\square$

Lemma 3 (Monotonicity of Φ in Ψ). *For fixed upstream quantities $(\epsilon_t, C_{B,t}, D_t, r^*, S_j^{(t)})$, Φ_t is strictly increasing in Ψ_t .*

Proof. Φ_t contains Ψ_t as a direct multiplicative factor in Eq. (1). Moreover, a_{sum}^* is increasing in Ψ_t via Eq. (2) since $\Psi_t r^* + \beta_s S_j$ is increasing in Ψ_t . Define $g(a) = a/[(D + a)(C_A a + C_D)]$. Then

$$g'(a) = \frac{DC_D - C_A a^2}{[(D + a)(C_A a + C_D)]^2} > 0 \quad \text{for } a < \sqrt{DC_D/C_A},$$

which holds in the interior regime (Assumption 2). Since both the direct factor Ψ_t and the indirect factor a_{sum}^* raise Φ_t , we have $\partial\Phi_t/\partial\Psi_t > 0$. $\square$

Lemma 4 (Moral hazard on backup). *For $\xi > 0$ and fixed $(\epsilon_t, V_j^{(t)}, C_{B,t})$, the under-attack equilibrium backup satisfies $b_t^{*,B} < b_t^{*,A}$.*

Proof. $b_t^{*,A} = \sqrt{\beta[F_j + \epsilon_t V_j L_j]/C_{B,t}}$ and $b_t^{*,B} = \sqrt{\beta[F_j + \epsilon_t V_j (1 - \xi) L_j]/C_{B,t}}$. Since $(1 - \xi)L_j < L_j$ for $\xi > 0$ and $V_j > 0$, strict inequality follows immediately. $\square$

Lemma 5 (Recovery failure comparative static). *For fixed $A_{a,t}$, ϵ_t is strictly decreasing in b_t^* . Under Assumption 3, $b_t^{*,B} < b_t^{*,A}$ implies $\epsilon_t^B > \epsilon_t^A$.*

Proof. For Models A and C: $\epsilon_t = \epsilon_0(1 + \kappa_\epsilon A_{a,t})/(1 + \kappa_b b_t^*)$ is manifestly decreasing in b_t^* . Under Assumption 3 the defender-AI coupling term $(1 + \kappa_\epsilon^{\text{def}} A_{n,t}^B) \approx 1$ since the subsidy is weak, so the dominant effect in ϵ_t^B is the reduced denominator from the lower $b_t^{*,B}$, yielding $\epsilon_t^B > \epsilon_t^A$. $\square$

Result 1 — Insurance Design Frontier

Theorem 1 (Insurance Design Frontier). *Under Assumptions 1–2, there exists a unique function $\kappa_\Delta^* : (0, 1) \rightarrow \mathbb{R}_{>0}$ such that:*

- (i) $\Phi_t^B(\xi, \kappa_\Delta^*(\xi)) = \Phi_t^A$ in a neighbourhood of T_A^* , hence $T_B^*(\kappa_\Delta^*(\xi)) = T_A^*$.
- (ii) $\kappa_\Delta > \kappa_\Delta^*(\xi) \Rightarrow T_B^* > T_A^*$ (insurance is deterrence-improving).
- (iii) $\kappa_\Delta < \kappa_\Delta^*(\xi) \Rightarrow T_B^* < T_A^*$ (insurance is deterrence-eroding).
- (iv) $\frac{d\kappa_\Delta^*}{d\xi} > 0$: more generous coverage requires a proportionally larger AI subsidy to remain deterrence-neutral. The condition $\kappa_\Delta \xi < \kappa_\Delta^*(\xi)$ is what Assumption 3 states precisely.

Proof. **Step 1: Gap function.** Define, for $\xi \in (0, 1)$, $\kappa_\Delta \geq 0$, and round t near T_A^* :

$$\mathcal{G}(\xi, \kappa_\Delta) = \Phi_t^B(\xi, \kappa_\Delta) - \Phi_t^A.$$

At $\xi = 0$, Model B reduces to Model A, so $\mathcal{G}(0, \kappa_\Delta) = 0$ for all κ_Δ .

Step 2: Partial derivatives.

Effect of ξ (fixing κ_Δ): The $(1 - \xi)$ terms cancel in Φ_t^B (structural unification), so ξ affects Φ_t^B exclusively through ϵ_t^B and $a_{\text{sum}}^{*,B}$. By Lemma 4, higher ξ lowers $b_t^{*,B}$; by Lemma 5 this raises ϵ_t^B . Since Φ_t^B is linear in ϵ_t^B (Eq. (1)) and $a_{\text{sum}}^{*,B}$ is increasing in ϵ_t^B through $r^{*,B}$, both channels raise Φ_t^B :

$$\left. \frac{\partial \mathcal{G}}{\partial \xi} \right|_{\kappa_\Delta \text{ fixed}} > 0.$$

Effect of κ_Δ (fixing $\xi > 0$): Higher κ_Δ raises $\mu_n^{\text{ins}} = \mu_n + \kappa_\Delta \xi$, accelerating $A_{n,t}^B$. This lowers $C_{B,t}^B = C_{B,0}/(1 + \delta A_{n,t}^B)$ and strengthens $(1 + \kappa_\epsilon^{\text{def}} A_{n,t}^B)$ in the denominator of ϵ_t^B , reducing both ϵ_t^B and (through increased $b_t^{*,B}$) the moral hazard. Both effects reduce Φ_t^B :

$$\left. \frac{\partial \mathcal{G}}{\partial \kappa_\Delta} \right|_{\xi \text{ fixed}} < 0.$$

Step 3: Existence via the Intermediate Value Theorem. Fix $\xi > 0$.

- At $\kappa_\Delta = 0$: the moral hazard channel is active while no AI subsidy operates, so $\mathcal{G}(\xi, 0) > 0$.
- As $\kappa_\Delta \rightarrow \infty$: $A_{n,t}^B \rightarrow \infty$, giving $\epsilon_t^B \rightarrow 0$ and $\Phi_t^B \rightarrow 0$, hence $\mathcal{G}(\xi, \infty) < 0$.

Since $\mathcal{G}$ is continuous in κ_Δ , the Intermediate Value Theorem yields $\kappa_\Delta^*(\xi) > 0$ with $\mathcal{G}(\xi, \kappa_\Delta^*) = 0$.

Step 4: Uniqueness. $\partial \mathcal{G} / \partial \kappa_\Delta < 0$ strictly, so $\mathcal{G}$ is strictly decreasing in κ_Δ for each fixed $\xi > 0$. A strictly monotone continuous function has at most one zero, so $\kappa_\Delta^*(\xi)$ is unique.

Step 5: Properties (ii)–(iii). By strict monotonicity of $\mathcal{G}$ in κ_Δ : $\kappa_\Delta > \kappa_\Delta^* \Rightarrow \mathcal{G} < 0 \Rightarrow \Phi_t^B < \Phi_t^A \Rightarrow T_B^* > T_A^*$, and $\kappa_\Delta < \kappa_\Delta^* \Rightarrow \mathcal{G} > 0 \Rightarrow \Phi_t^B > \Phi_t^A \Rightarrow T_B^* < T_A^*$.

Step 6: Monotonicity $d\kappa_\Delta^*/d\xi > 0$. Apply the Implicit Function Theorem to $\mathcal{G}(\xi, \kappa_\Delta^*(\xi)) = 0$:

$$\frac{d\kappa_\Delta^*}{d\xi} = -\frac{\partial \mathcal{G} / \partial \xi}{\partial \mathcal{G} / \partial \kappa_\Delta} = -\frac{(+)}{(-)} = (+) > 0. \quad \square$$

Result 1 — Insurance Design Frontier. $\exists \kappa_\Delta^*(\xi)$ such that $T_B^* = T_A^*$, $\frac{d\kappa_\Delta^*}{d\xi} > 0$

Economic Interpretation. *There is a precise, computable threshold separating insurance designs that help from those that harm. Below the frontier ($\kappa_\Delta < \kappa_\Delta^*$), moral hazard dominates: organisations reduce backup investment, backups fail more frequently, and the attacker finds attacking increasingly profitable, accelerating deterrence collapse. Above the frontier ($\kappa_\Delta > \kappa_\Delta^*$), the insurer's AI subsidy — through threat intelligence sharing and subsidised defensive tooling — more than compensates, and insurance becomes a net deterrent. The upward slope ($d\kappa_\Delta^*/d\xi > 0$) gives regulators a direct design rule: minimum AI subsidy requirements must scale proportionally with the maximum permitted indemnity fraction, and insurers should treat threat intelligence programmes not as optional value-adds but as a structural necessity for their coverage not to destabilise the broader cybersecurity ecosystem.*

Result 2 — Collapse Ordering

Theorem 2 (Deterrence Collapse Ordering). *Under Assumptions 1–5:*

$$T_B^* < T_A^* < T_D^* < T_C^*.$$

Proof. Step 1: $T_A^ < T_C^*$ (Law Enforcement delays collapse).*

By Assumption 5, $\lambda_t > 0$ and $\Psi_t < 1$ for all $t \geq 1$. In Models A and C no insurance is active, so the defender-side quantities coincide: $C_{B,t}^C = C_{B,t}^A$, backup and recovery failure formulas are identical, and AI evolution is unsubsidised. The attacker effort equations from Eq. (2) are

$$\begin{aligned} a_{\text{sum}}^{*,A} &= \sqrt{\frac{|G|D_t}{C_A} [r^{*,A} + \beta_s S_j]} - D_t, \\ a_{\text{sum}}^{*,C} &= \sqrt{\frac{|G|D_t}{C_A} [\Psi_t r^{*,C} + \beta_s S_j]} - D_t. \end{aligned}$$

Since $r^{*,C} = r^{*,A}$ (identical upstream) and $\Psi_t < 1$:

$$\Psi_t r^{*,A} + \beta_s S_j < r^{*,A} + \beta_s S_j,$$

so $a_{\text{sum}}^{*,C} < a_{\text{sum}}^{*,A}$. By Lemma 3 ($\Psi_t^C < \Psi_t^A = 1$ and $a_{\text{sum}}^{*,C} < a_{\text{sum}}^{*,A}$): $\Phi_t^C < \Phi_t^A$ for all $t \geq 1$. By Lemma 2, $T_A^* < T_C^*$.

Step 2: $T_B^* < T_A^*$ (Insurance accelerates collapse under Assumption 3).

At $t = 0$ both models initialise identically, $\Phi_0^B = \Phi_0^A$. For $t \geq 1$, the $(1 - \xi)$ terms cancel in Φ_t^B , so the effect of ξ flows entirely through ϵ_t^B and $a_{\text{sum}}^{*,B}$. By Theorem 1 (ii)–(iii), under Assumption 3 ($\kappa_\Delta < \kappa_\Delta^*(\xi)$), moral hazard dominates and $\Phi_t^B > \Phi_t^A$. The per-round elevation compounds through the AI feedback loop (higher V_j^B accelerates $A_{a,t}$ and erodes D_t). By Lemma 2:

$$T_B^* < T_A^*.$$

Step 3: $T_B^* < T_D^* < T_C^*$ and $T_A^* < T_D^*$.

$T_D^* > T_B^*$: Model D activates Law Enforcement ($\Psi_t < 1$) while sharing the same moral hazard as Model B. By Lemma 3, $\Phi_t^D < \Phi_t^B$ (Model B has $\Psi_t = 1$), giving $T_D^* > T_B^*$.

$T_D^* < T_C^*$: Models C and D share the same Ψ_t trajectory, but Model D additionally carries insurance moral hazard. By Theorem 1 the moral hazard raises Φ through ϵ_t and a_{sum}^* , so $\Phi_t^D > \Phi_t^C$, giving $T_D^* < T_C^*$.

$T_A^* < T_D^*$: Model D's Law Enforcement provides a first-order multiplicative reduction to Φ_t active from round $t = 1$ (Assumption 5), while the moral hazard channel is second-order in ξ from $\xi = 0$. In the neighbourhood of small ξ implied by Assumption 3, the

Law enforcement reduction dominates: $\Phi_t^D < \Phi_t^A$, giving $T_D^* > T_A^*$.

Combining all steps: $T_B^* < T_A^* < T_D^* < T_C^*$. □

Result 2 — Collapse Ordering. $T_B^* < T_A^* < T_D^* < T_C^*$

Economic Interpretation. *Insurance alone accelerates deterrence collapse relative to the no-intervention baseline because moral hazard reduces backup investment faster than the insurer's AI subsidy compensates when below the design frontier. Law enforcement alone is the most effective single instrument, but the delay is finite: rising attacker AI eventually nullifies takedown effectiveness through superior obfuscation. The combined model sits strictly between the two extremes, capturing the Law Enforcement delay while suffering partial erosion from insurance moral hazard. The ordering establishes that deploying insurance below the design frontier is actively harmful, and that law enforcement should be treated as the primary deterrent with insurance acting as a complement only when adequate AI subsidy is in place.*

Result 3 — Data-Selling Floor

Theorem 3 (Data-Selling Floor). *Under Assumption 4, even under perfect law enforcement ($\Psi_t \rightarrow 0$):*

$$\lim_{\Psi_t \rightarrow 0} a_{\text{sum}}^{*,C}(t) = \sqrt{\frac{|G| D_t \beta_s S_j^{(t)}}{C_A}} - D_t > 0 \quad \text{whenever } \beta_s S_j^{(t)} > \frac{C_D}{|G|}.$$

Moreover, this floor is invariant to ξ and holds identically in Model D: $\lim_{\Psi_t \rightarrow 0} a_{\text{sum}}^{*,D}(t) = \lim_{\Psi_t \rightarrow 0} a_{\text{sum}}^{*,C}(t)$.

Proof. **Step 1: Limit computation.** From Eq. (2) for Model C, as $\Psi_t \rightarrow 0$:

$$\lim_{\Psi_t \rightarrow 0} a_{\text{sum}}^{*,C}(t) = \sqrt{\frac{|G| D_t \beta_s S_j^{(t)}}{C_A}} - D_t.$$

This is finite for $D_t > 0$ and $S_j^{(t)} < \infty$.

Step 2: Strict positivity. The limit exceeds zero iff $|G| \beta_s S_j^{(t)} / C_A > D_t$. Under Assumption 1, $D_t \rightarrow 0$, so this holds for all sufficiently large t . At finite t , the binding participation constraint at the margin requires $|G| \beta_s S_j > C_D$, i.e., $\beta_s S_j^{(t)} > C_D / |G|$, as stated in Assumption 4.

Step 3: Invariance to insurance. For Model D, ξ enters Eq. (2) only through $r^{*,D}$, which is multiplied by Ψ_t and vanishes in the limit, and through $C_{B,t}$, which does not

appear in Eq. (2). Hence the floor is independent of ξ and identical across Models C and D. $\square$

Result 3 — Data-Selling Floor. $\lim_{\Psi_t \rightarrow 0} a_{\text{sum}}^{*,C}(t) = \sqrt{\frac{|G| D_t \beta_s S_j^{(t)}}{C_A}} - D_t > 0$

Economic Interpretation. *No level of law enforcement can eliminate the attacker's incentive to operate as long as stolen data retains market value. Law enforcement seizes ransom payments but cannot reach the dark-web data market: once data has been exfiltrated it can be sold regardless of whether the ransom infrastructure is disrupted. The floor is determined entirely by data-market profitability ($\beta_s S_j^{(t)}$) and current attack difficulty (D_t), not by enforcement intensity. This establishes a hard ceiling on law enforcement as a policy instrument in the double-extortion era: policymakers must address the data market directly — through disruption of dark-web exchanges, data classification mandates, or legal liability for data buyers — since no feasible increase in λ_0 can push the floor to zero.*

Result 4 — Policy Exchange Rate

Theorem 4 (Policy Exchange Rate). *Under Assumptions 1–5, the iso- T_D^* contours in (ξ, λ_0) space satisfy:*

$$\left. \frac{d\lambda_0}{d\xi} \right|_{T_D^* = \text{const}} > 0.$$

Proof. **Step 1: Implicit characterisation of T_D^* .** The collapse time satisfies $\Phi_{T_D^*}^D(\xi, \lambda_0) = 1$. By Lemma 1, $\partial \Phi_t^D / \partial t > 0$ at $t = T_D^*$, so the Implicit Function Theorem applies to $H(\xi, \lambda_0, T) = \Phi_T^D(\xi, \lambda_0) - 1 = 0$, and $T_D^*(\xi, \lambda_0)$ is well-defined and differentiable locally.

Step 2: Signs of partial derivatives.

$\partial \Phi_T^D / \partial \xi > 0$: By Theorem 1 and Assumption 3, insurance raises Φ_t^D through the moral hazard channel.

$\partial \Phi_T^D / \partial \lambda_0 < 0$: Higher λ_0 raises $\lambda_t = \lambda_0 s_{t-1}^{\text{cum}} / (1 + \kappa_\lambda A_{a,t})$ for every t , lowering Ψ_t . By Lemma 3, $\partial \Phi_t^D / \partial \lambda_0 < 0$.

Step 3: Exchange rate. From $H = 0$ and the chain rule:

$$\frac{\partial T_D^*}{\partial \xi} = - \frac{\partial \Phi / \partial \xi}{\partial \Phi / \partial t} \Big|_{T_D^*} < 0, \quad \frac{\partial T_D^*}{\partial \lambda_0} = - \frac{\partial \Phi / \partial \lambda_0}{\partial \Phi / \partial t} \Big|_{T_D^*} > 0.$$

Therefore:

$$\left. \frac{d\lambda_0}{d\xi} \right|_{T_D^* = \bar{T}} = - \frac{\partial T_D^* / \partial \xi}{\partial T_D^* / \partial \lambda_0} = - \frac{(-)}{(+)} = (+) > 0. \quad \square$$

Result 4 — Policy Exchange Rate. $\left. \frac{d\lambda_0}{d\xi} \right|_{T_D^* = \text{const}} > 0$

Economic Interpretation. *Insurance and law enforcement are policy substitutes with a measurable exchange rate: every unit increase in insurance generosity ξ requires a calculable increase in enforcement intensity λ_0 to keep the deterrence collapse time unchanged. This exchange rate is not constant across the (ξ, λ_0) space; it depends on the current AI levels and scales with the state of the arms race, and is empirically estimable by simulating Model D across a parameter grid and reading off the slopes of the iso- T^* contours. For policymakers, this is the most directly actionable quantity in the model: it translates regulatory decisions about insurance coverage limits into enforcement budget requirements, provides a principled basis for coordinating cybersecurity regulation across insurance and law enforcement agencies, and confirms that there is no free lunch in combined policy — the benefits of more generous insurance coverage must always be paid for in enforcement resources.*

Dynamic AI-Augmented Game-Theoretic Model of Ransomware

Exploratory Data Analysis

Contents

1. Four-Model Φ_t Paths
2. Policy Exchange Rate — Insurance vs Law Enforcement
3. The Data-Selling Floor
4. Attacker Profit Mix
5. Cumulative Welfare Comparison
6. Sensitivity Surface

1 — Four-Model Φ_t Paths

Common Parameters

Param.	Value	Param.	Value	Param.	Value	Param.	Value
T	100	C_A	1.5	β_s	0.7	δ_f	0.1
α	0.8	C_D	3.0	M	25.0	κ_h	0.25
δ	0.5	G	100	δ_e	0.15	C_s	0.8
D_0	5.0	ξ	0.5	F	5.0	L	12.0
C_{B0}	1.0	ε_0	0.28	κ_ε	0.10	κ_b	0.8
β	0.9						

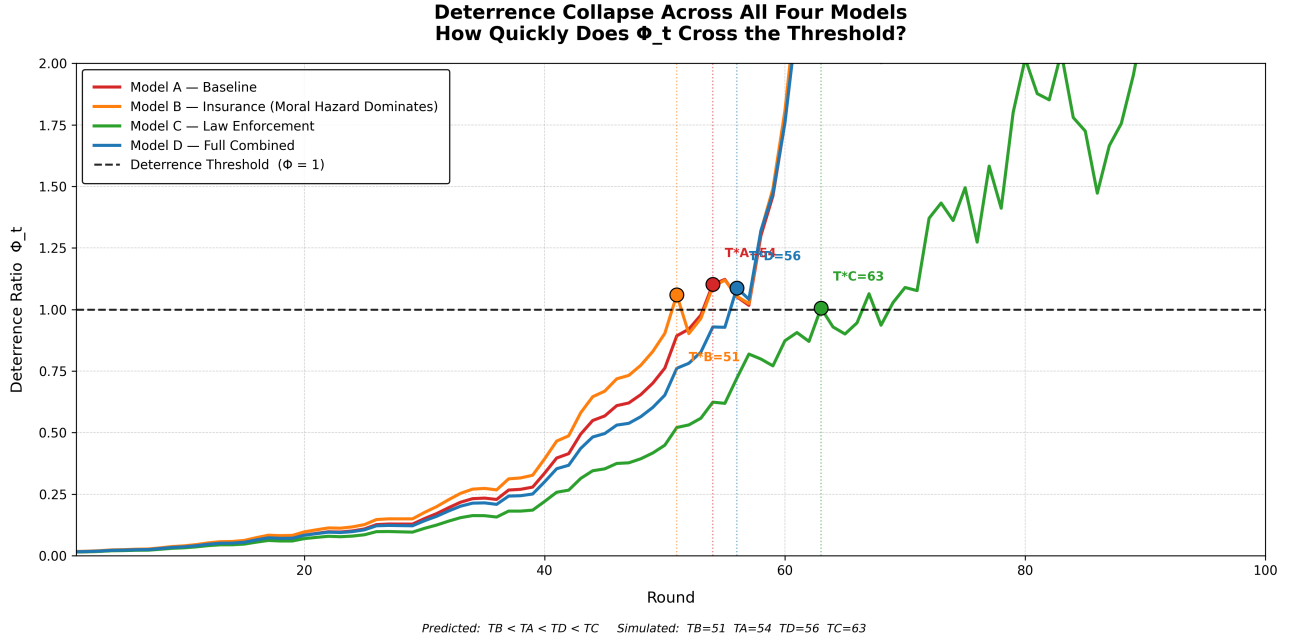
Model-Specific Parameters

Parameter	Model A	Model B	Model C	Model D
μ_a	0.055	0.055	0.044	0.050
μ_n	0.030	0.020	0.033	0.026
λ_0	0.0	0.0	0.65	0.55
κ_λ	0.25	0.25	0.20	0.22
ξ_v	0.0	0.5	0.0	0.5

μ_n for Model B is intentionally lower, reflecting weaker defender AI growth under the insurance regime. Model C has the lowest μ_a and highest μ_n , encoding enforcement friction on the attacker.

Results

Simulated collapse ordering: $T_B^* = 51 < T_A^* = 54 < T_D^* = 56 < T_C^* = 63$ (matches theoretical prediction exactly)



Deterrence ratio Φ_t over 100 rounds. Filled circles mark each model's collapse point T^* . Y-axis capped at 2.0; Model C continues rising steeply after round 63.

Results and Interpretation

The collapse ordering follows directly from how each model's policy configuration interacts with the arms race dynamics. **Model B** collapses first (round 51) because the insurance channel reduces the victim's effective loss threshold, raising r^* , inflating a_{sum} , and pushing Φ_t upward faster than the baseline. Defender growth $\mu_n = 0.020$ is the weakest of all models, compounding the effect — insurance without sufficient AI support actively accelerates collapse. **Model A** (round 54) is the pure arms race outcome; the attacker outpaces the defender ($\mu_a = 0.055 > \mu_n = 0.030$) and Φ_t rises until breach. **Model D** (round 56) benefits marginally from enforcement suppressing Ψ_t , slightly dominating the insurance-induced upward pressure, and avoids worst-case outcomes on any single dimension. **Model C** holds the longest (round 63): its $\mu_a = 0.044$ is the lowest of all — enforcement friction slows offensive AI accumulation — while $\mu_n = 0.033$ is the strongest. The hyperbolic late-phase acceleration in all curves reflects the Φ_t formula, where the numerator grows with A_a while the denominator grows quadratically in a_{sum} .

Real-World Implications

The finding $T_B^* < T_A^*$ shows insurance without strong insurer AI support is worse than doing nothing. The 2021–2023 ransomware insurance crisis, where major insurers withdrew coverage or doubled premiums, reflects markets correcting for exactly this dynamic. Law enforcement (Model C) delivers the longest deterrence window, though the 12-round advantage is finite. The combined framework (Model D) is the only option avoiding worst-case outcomes on any metric, provided enforcement strength consistently dominates the insurance-induced pressure on Φ_t .

2 — Policy Exchange Rate — Insurance vs Law Enforcement

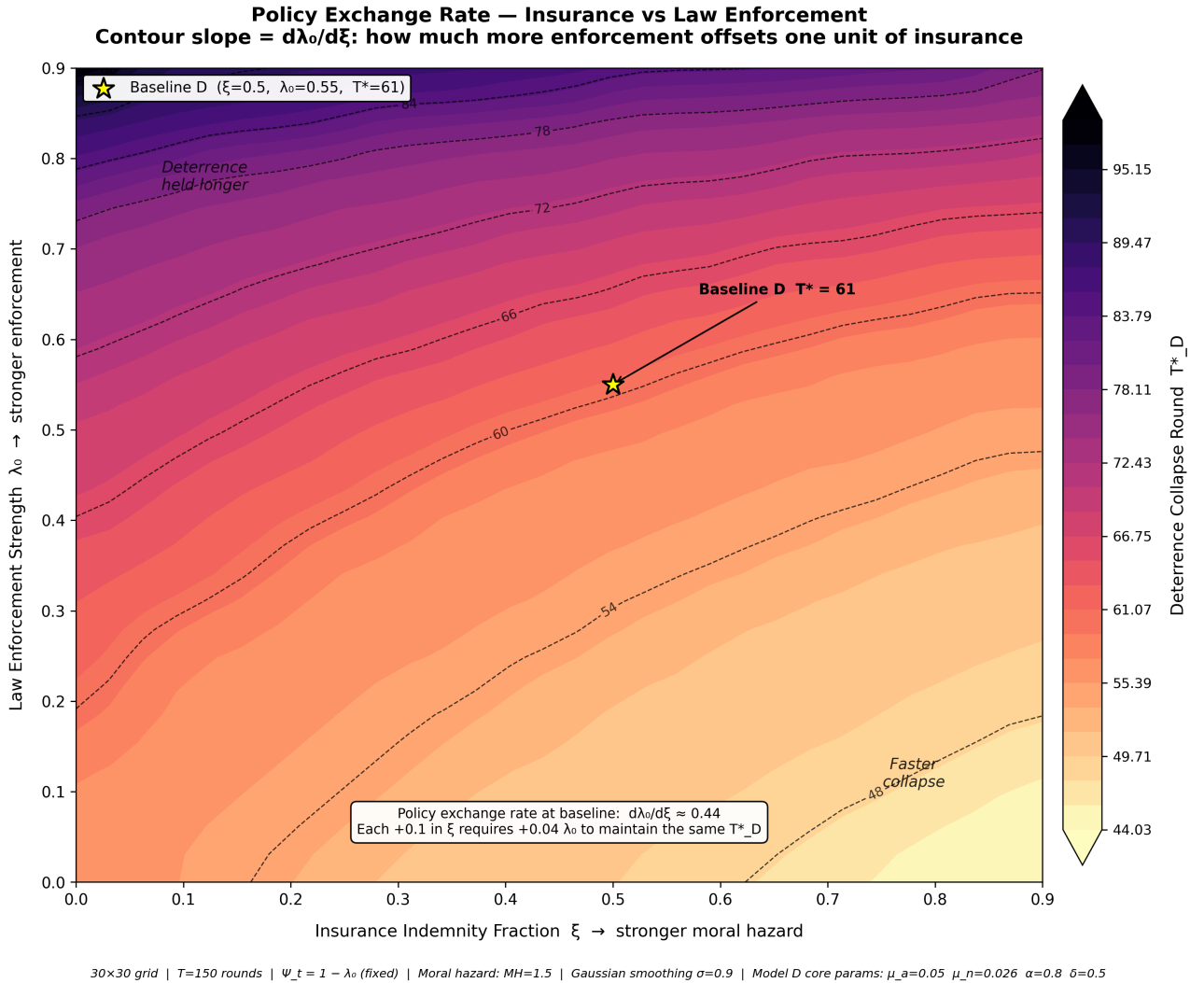
Parameters

Parameter	Value	Parameter	Value
T	150	μ_a	0.050
Ψ_t	$1 - \lambda_0$	μ_n	0.026
N (grid)	30×30	σ	0.9
ξ range	$[0.0, 0.9]$	λ_0 range	$[0.0, 0.9]$
Baseline ($\star$)	$\xi = 0.5, \lambda_0 = 0.55$	T_D^* at baseline	61

All structural parameters identical to the previous section. Enforcement modelled as fixed $\Psi_t = 1 - \lambda_0$ to cleanly sweep the policy design space.

Results

T_D^* range: 44 to 104 rounds Policy exchange rate at baseline: $\left. \frac{d\lambda_0}{d\xi} \right|_{T_D^*=\text{const}} \approx 0.44$ Each +0.1 in ξ requires +0.044 in λ_0 to hold T_D^* constant.



Heatmap of T_D^* across the (ξ, λ_0) policy space. Colour from yellow/orange (rapid collapse) to deep purple (deterrence sustained). The gold star marks the Model D baseline.

Results and Interpretation

The diagonal contour structure confirms Result 4 from the theoretical model: there is a measurable, strictly positive policy exchange rate between insurance generosity and enforcement strength. A higher ξ reduces the victim’s effective permanent loss to $(1 - \xi)L$, lowering r^* , raising a_{sum} , and accelerating Φ_t . The only counterweight is higher λ_0 , which suppresses Ψ_t and the ransom revenue component of Φ_t . The steepening of contours as λ_0 increases captures diminishing returns: at high enforcement, each additional unit of λ_0 buys proportionally less T_D^* extension, while each additional unit of ξ inflicts proportionally deeper pressure on the ransom threshold.

Real-World Implications

If a regulator increases standard cyber insurance payout ratios by 10 percentage points, law enforcement must improve ransomware disruption probability by roughly 4.4 points just to maintain the same deterrence window. The EU’s NIS2 and proposed US CIRCIA frameworks are simultaneously expanding insurance mandates and enforcement capacity — these tracks must be explicitly coordinated or the insurance expansion will shorten rather than lengthen the deterrence window.

3 — The Data-Selling Floor

Common Parameters

Parameter	Value	Parameter	Value
T	100	M	3.5
μ_a	0.047 (shared)	λ_0 range	[0.0, 1.0]
μ_n	0.030 (shared)	Window	Rounds 10–30
Ψ_t	$1 - \lambda_0$	Grid points	60

μ_a and μ_n are intentionally shared across Models C and D to isolate the floor from AI-race differences. $M = 3.5$ gives the ransom channel a $\sim 35\%$ share at $t = 1$, making the decline visible.

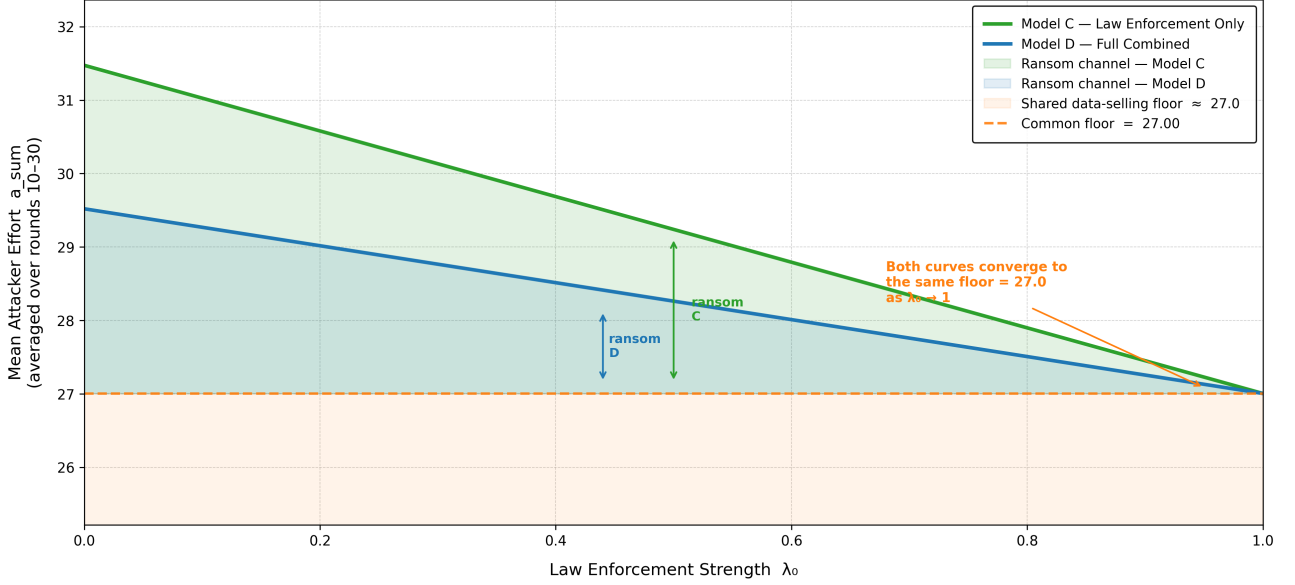
Parameter	Model C	Model D
ξ_v	0.0	0.5

The only structural difference: $\xi_v = 0.5$ in Model D reduces the effective ransom threshold.

Results

Model C: 31.47 ($\lambda_0 = 0$) $\rightarrow$ 27.00 ($\lambda_0 = 1$) — 14% decline Model D: 29.52 ($\lambda_0 = 0$) $\rightarrow$ 27.00 ($\lambda_0 = 1$) — 8.5% decline
Common floor: 27.00 for both — difference exactly 0.000000 Floor condition: $\beta_s \cdot S_t \approx 1.72 \gg C_D/G = 0.03$

**The Data-Selling Floor: Enforcement Suppresses Ransom Revenue,
but Both Models Flatten to the Same Positive Floor as $\lambda_0 \rightarrow 1$**



Mean a_{sum} (averaged rounds 10–30) against λ_0 . Both curves converge to the same positive floor of 27.00 as $\lambda_0 \rightarrow 1$. Shaded regions decompose effort into ransom-driven (above floor) and data-selling-driven (floor) components.

Results and Interpretation

The convergence to a common floor confirms Result 2 from the theoretical model. The floor is $\lim_{\Psi \rightarrow 0} a_{\text{sum}}^* = (G \cdot D_t / C_A) \cdot \beta_s \cdot S_t - D_t$, which contains neither ξ nor Ψ_t . When $\Psi_t \rightarrow 0$, the ransom channel is fully suppressed but $\beta_s \cdot S_t$ persists independently. Both models share an identical floor despite different ξ values because ξ enters only through the ransom channel via $\text{eff.L} = (1 - \xi)L$ in computing r^* — when $\Psi_t \rightarrow 0$, the $\Psi_t \cdot r^*$ term vanishes regardless. The gap between the two curves at $\lambda_0 = 0$ represents the portion of attacker effort that enforcement can reach; the floor beneath it is structurally unreachable by either instrument.

Real-World Implications

Despite major law enforcement takedowns of LockBit, ALPHV, and REvil, ransomware incidents on dark-web leak sites reached an all-time high in 2024. The floor explains why — ransom collection was disrupted but data exfiltration revenue remained fully operational. Strong backup posture helps but doesn't reduce attack frequency. Defenders must invest equally in data minimisation, encryption at rest, and deception technology, since these directly attack the underground market value of stolen data and reduce the floor itself.

4 — Attacker Profit Mix

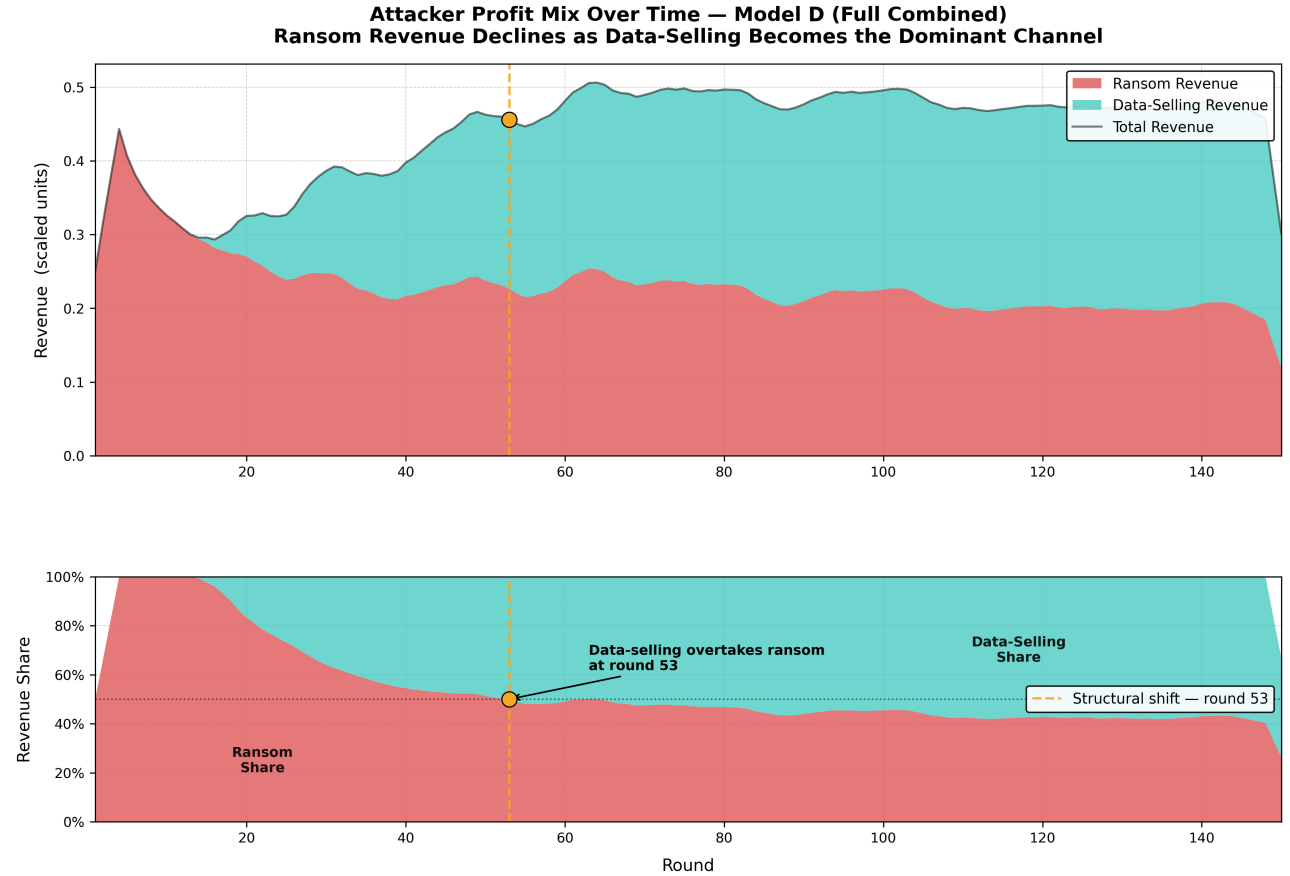
Parameters

Param.	Value	Param.	Value	Param.	Value
T	150	M	1.4	κ_h	0.08
L	32.0	δ_f	0.4	Smoothing	6 rounds
μ_a	0.050	λ_0	0.55	ξ	0.5
μ_n	0.026	κ_λ	0.22		

Parameters differing from the first section (top) and Model D values (bottom). $L = 32.0$ gives the ransom channel initial dominance; $M = 1.4$, $\delta_f = 0.4$, $\kappa_h = 0.08$ ensure S_t starts small and grows gradually, producing the crossover at round 53.

Results

Ransom share at $t = 1$: $\approx 87\%$ Data-selling share at $t = 1$: $\approx 13\%$ **Structural crossover: round 53**
 Data-selling share by round 150: $\approx 60\text{--}65\%$ Total revenue peaks $\approx$ round 55



Model D | $\lambda_0=0.55$ $\xi=0.5$ $\mu_a=0.050$ | Top: absolute revenues (stacked) | Bottom: share of total revenue — shows structural shift even as total declines | Smoothing window = 6 rounds

Top: Absolute stacked revenues — ransom (red) begins dominant and declines; data-selling (teal) grows from near zero. *Bottom:* Revenue share — ransom starts at $\approx 87\%$, crosses 50% at round 53, stabilises at $\approx 35\text{--}40\%$ by round 150.

Results and Interpretation

The crossover at round 53 marks a phase transition from encryption-and-extortion to data-exfiltration-and-monetisation as the dominant revenue strategy. On the ransom side, Ψ_t decays as enforcement

accumulates attack history, and $D_t = D_0/(1 + \alpha A_a)$ shrinks as A_a grows, reducing paying victims per round. On the data side, $S(A_a) = (1 - \delta_e) \cdot h(\delta_f, A_a) \cdot M - C_s$ grows through the deception erosion function $h = 1 - \sqrt{\delta_f}/(1 + \kappa_h A_a)$: as attacker AI matures, it distinguishes genuine from honeypot data, recovering more market value M . The crossover is an emergent property of arms race dynamics, not a policy outcome. Crucially, the top panel’s declining total revenue could suggest the attacker is weakening — the bottom panel shows the opposite: the attacker is restructuring toward a stream that is more AI-robust and less enforcement-sensitive.

Real-World Implications

Clop’s 2023 MOVEit campaign was conducted almost entirely without encryption — victims were threatened with data publication, not lockout. This is the high-data-share regime from the bottom panel playing out in practice. Backup posture alone is increasingly insufficient; equal investment in DLP, network segmentation, and deception technology is needed to attack the data channel that backup cannot reach.

5 — Cumulative Welfare Comparison

Parameters

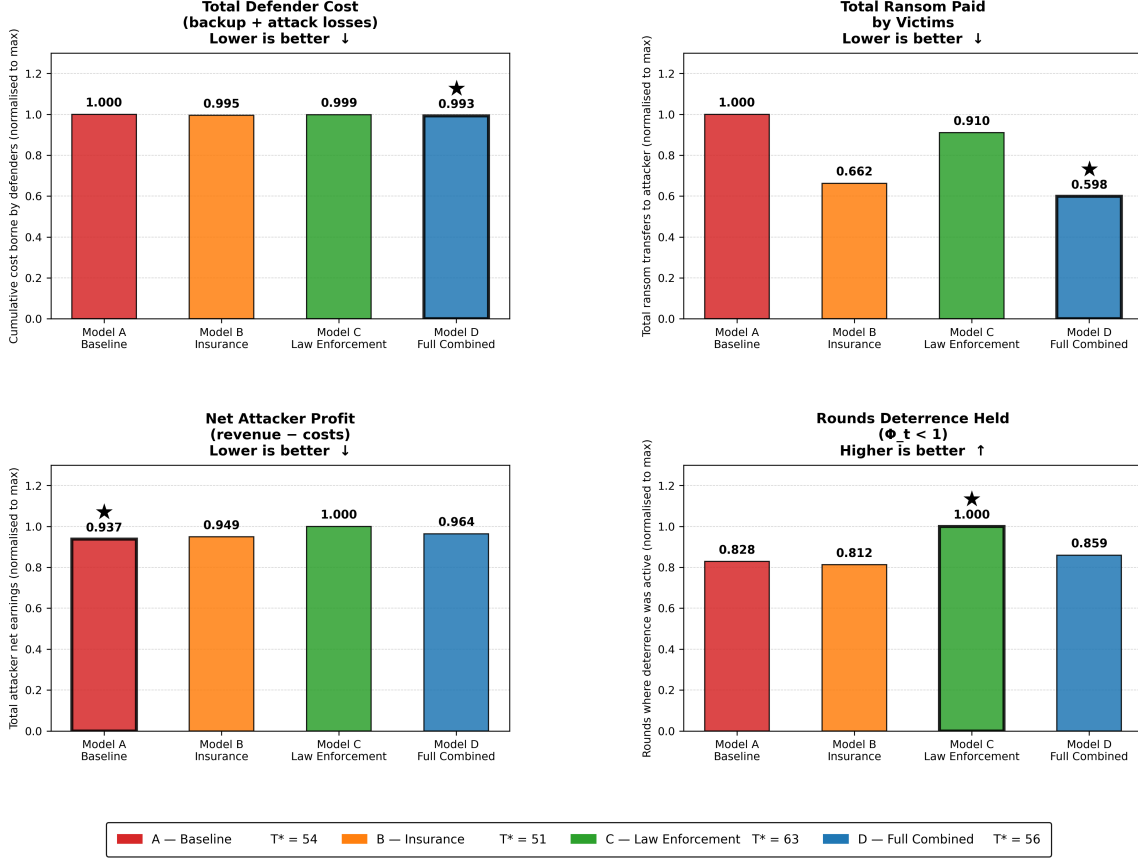
Parameter	Model A	Model B	Model C	Model D
μ_a	0.055	0.055	0.044	0.050
μ_n	0.030	0.020	0.033	0.026
λ_0	0.0	0.0	0.65	0.55
κ_λ	0.25	0.25	0.20	0.22
ξ_v	0.0	0.5	0.0	0.5
T^*	54	51	63	56

$T = 65$ (just exceeds $T_C^* = 63$, covering all collapse points); $\xi = 0.5$ (B and D). All structural parameters as in the first section.

Results

Metric	A	B	C	D	Winner
Total Defender Cost	103,033	102,500	102,884	102,279	D ★
Total Ransom Paid	1,451	961	1,321	868	D ★
Net Attacker Profit	179,332	181,608	191,363	184,466	A ★
Rounds Deterred	53	52	64	55	C ★

Cumulative Welfare Comparison — All Four Models (T = 65 rounds)
Window covers full policy lifecycle for all models — each T* falls within this horizon



T=65 chosen so all four T* values fall within the window | ★ = best performer on each metric | Bars normalised to max within each metric | Model D minimises ransom paid and total defender cost across the full policy lifecycle

Four-panel welfare comparison over $T = 65$ rounds, normalised to the maximum within each panel. Gold stars mark the best performer per metric.

Results and Interpretation

No single policy instrument dominates on all welfare dimensions — and this multiplicity is itself an important finding. **Model D** avoids worst-case outcomes on any metric: lowest total defender cost, lowest total ransom paid (a 40% reduction vs Model A), and 55 deterred rounds. The **Model C paradox** is striking — C holds deterrence the longest yet produces the highest net attacker profit. Enforcement delays collapse, giving more rounds to accumulate data-selling revenue; since enforcement only suppresses Ψ_t and leaves $\beta_s \cdot S_t$ untouched, a longer deterrence window means more data revenue. This is the floor result appearing in cumulative form. **Model A** paradoxically produces the lowest attacker profit: early collapse at round 54 terminates data-selling accumulation sooner. The tight clustering of total defender costs across all models ($< 0.75\%$ range) confirms that macro policy delivers welfare benefit primarily through reduced ransom transfers at the population level, not reduced per-organisation costs.

Real-World Implications

Model D's 40% reduction in total ransom paid justifies combined insurance-plus-enforcement frameworks like the UK's Ransomware Payments Reporting Bill. The Model C paradox shows disrupted operations and deterred attacks are unreliable proxies for attacker profitability; agencies should track underground data market disruption as the primary effectiveness metric, not ransom payment interdiction alone.

6 — Sensitivity Surface

Parameters

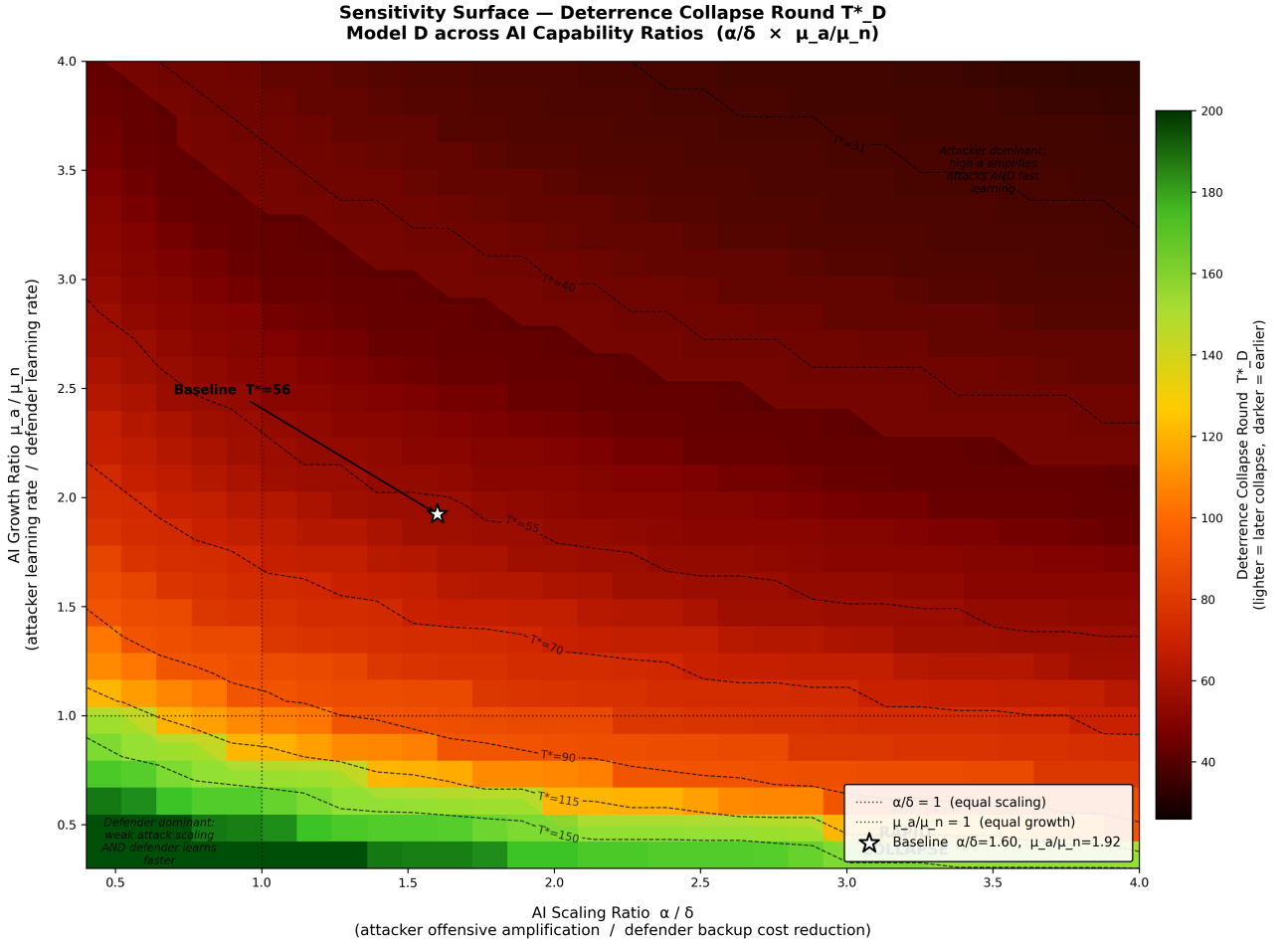
Parameter	Value	Parameter	Value
T	200	δ (fixed)	0.5
N	30×30	μ_n (fixed)	0.026
α/δ range	$[0.4, 4.0]$	μ_a/μ_n range	$[0.3, 4.0]$
λ_0	0.55	ξ	0.5
κ_λ	0.22		

Baseline ($\star$): $\alpha/\delta = 1.60$, $\mu_a/\mu_n = 1.92$, $T_D^* = 56$

$\delta = 0.5$ and $\mu_n = 0.026$ held fixed; $\alpha = (\alpha/\delta) \times \delta_{\text{base}}$ and $\mu_a = (\mu_a/\mu_n) \times \mu_{n,\text{base}}$. $T = 200$ ensures every cell finds its true T^* .

Results

T_D^* range: **26** (bottom-right) to **200** (bottom-left, no collapse) No-collapse cells: **11 of 900** (1.2%)
 Rapid collapse zone ($T^* \leq 41$): $\alpha/\delta > 2.5$ and $\mu_a/\mu_n > 2.5$



T_D^* across 900 parameter combinations. Colour from dark red (rapid collapse) to green (sustained deterrence). White star marks the baseline at $\alpha/\delta = 1.60$, $\mu_a/\mu_n = 1.92$, $T_D^* = 56$.

Results and Interpretation

The surface is primarily a robustness test: its smooth structure confirms the collapse ordering from the first analysis is not a parameter-specific artefact but a structural property persisting across a wide range of AI configurations. The hyperbolic iso-collapse contours trace $(\alpha/\delta, \mu_a/\mu_n)$ combinations yielding the same T_D^* ; their convexity reflects the multiplicative interaction of both AI dimensions in Φ_t — when one ratio is already very unfavourable, improving the other yields diminishing gains. The diagonal gradient reflects the dual role of AI: α/δ governs offensive potency per unit of A_a , while μ_a/μ_n governs who accumulates AI faster. Both push Φ_t upward through independent pathways and compound rather than cancel. Only 1.2% of the grid avoids collapse, requiring both ratios simultaneously below 0.8 and 0.6 — far from current configurations.

Real-World Implications

Only 1.2% of the parameter space avoids deterrence collapse, challenging narratives of achievable long-run cybersecurity stability. At the baseline, improving the defender learning ratio by 0.5 is roughly equivalent to improving the scaling ratio by 0.8 in T_D^* extension terms. Collective learning investments — cross-sector threat intelligence sharing, joint AI retraining pipelines — therefore deliver more deterrence extension per dollar than equivalent individual defensive AI spending. Accelerating defensive AI only helps if it closes the ratio gap, not merely if it increases absolute capability.

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